Manuscript: Post-transcriptional spliceosome accumulation marks a writer-dominant malignant cell state in glioblastoma

This document describes all supplementary figures and tables.

Figure S1 | CopyKAT purity control. (A) Aneuploid (tumor) fraction per sample estimated by copy-number inference (18 samples, 16,255 cells called). (B) WEBS distribution across LLS/LDTS/LHS within aneuploid-only cells (n = 4,421; Kruskal?Wallis p < 10?¹?), demonstrating that WEBS stratification is not an artifact of non-malignant admixture.

Figure S2 | Parameter sensitivity. (a) WEBS computed at UCell maxRank from 5% to 90% of genes vs the default (1,500 genes); WEBS is stable up to maxRank ? 0.10 and drifts at larger values, motivating the fixed maxRank = 1,500 setting. (b) Leave-one-out and leave-two-out removal of writer/eraser genes (57 perturbations; median Spearman ? = 0.95 vs default). (c) Cross-platform consistency of patient-level WEBS between Smart-seq2 and 10x in GSE131928 (? = 0.67, n = 10 shared patients).

Figure S3 | Method benchmarking and computational performance. Patient-level WEBS vs AUCell (? = 0.95), ssGSEA (? = 0.87), all-gene mean (? = 0.92); PLAGE is sign-inverted (? = ?0.97, a known PCA orientation artifact). Runtime and memory scaling at 1k?35.8k cells: WEBS completes 35,823 cells in 213 s with <2 MB incremental memory; ssGSEA exceeds available memory beyond ~25k cells.

Figure S4 | Pan-cancer scope. WEBS?survival associations across TCGA cohorts (GBM, LUAD, BRCA, SKCM): significant in LUAD (HR = 0.64, p = 0.039) but not in BRCA/SKCM/GBM-bulk, delineating the cancer-type-specific scope of the axis.

Figure S5 | Bootstrap stability of the LASSO-Cox feature selection. Selection frequency of candidate genes across 1,000 bootstrap resamples of the training cohort; HDAC1 (99.9%), LDHB (98%), VEGFA (94%), KAT8 (89%) are the most stable features.

Figure S6 | SHAP interpretability of the six-gene model. Mean |SHAP| per gene overall and by risk group; HDAC1 is the dominant contributor (protective direction, consistent with its eraser role).

Figure S7 | Spliceosome specificity control. Layer discordance (protein ? ? mRNA ? vs WEBS) for the spliceosome (0.81) against 10,000 random gene sets matched for proteome abundance (empirical p = 0.22), demonstrating that partial discordance is generic to abundant complexes; spliceosome specificity therefore rests on direction-uniform up-regulation and functional evidence rather than discordance magnitude. The proteasome is shown as a secondary housekeeping-complex case (0.77).

Figure S8 | Spliceosome acetylation-site heatmap. Top 40 differentially acetylated spliceosome sites (limma, FDR < 0.05), samples ordered by WEBS group; site-level acetylation is enriched in the writer-dominant state (OR = 7.36, p = 1.3 × 10?¹?).

Figure S9 | MOFA2 four-view integration (CPTAC). (a) Factor?WEBS Spearman correlations across 10 latent factors; Factor 2 covaries with WEBS (? = ?0.637, p < 10?¹²). (b) Factor 2 vs protein-level WEBS across 99 CPTAC-GBM samples.

Figure S10 | Spatial autocorrelation of WEBS. Moran's I per Visium slice with 1,000-permutation p-values; significant clustering in GBM3, GBM5_2, and GBM4.

Figure S11 | GSE83300 external validation. Kaplan?Meier for the six-gene signature in GSE83300 (n = 50, 39 events): HR = 2.76 with concordant direction but p = 0.74, consistent with limited power at this sample size; reported transparently.

Figure S12 | Risk × treatment interaction. ?log10(p) of the signature-by-treatment interaction term in Cox models (CGGA-693 and CGGA-325), for temozolomide and radiotherapy. Significant interactions in both cohorts indicate treatment-context-dependent prognostic value.

Figure S13 | Joint MGMT × signature stratification. Kaplan?Meier curves of four strata (MGMT methylated/unmethylated × high/low risk) in CGGA-325 and CGGA-693 (both $p < 10?^{13}$).

Figure S14 | GLASS longitudinal WEBS. WEBS in paired primary and first-recurrence glioma samples (n = 179 pairs; paired Wilcoxon p = 0.09): no systematic change at recurrence.

Figure S15 | Meta-analysis of the six-gene signature. Random-effects meta-analysis across TCGA-GBM, CGGA-693, and CGGA-325 (primary pooled HR = 1.35, p = 0.0028; $I^2 = 91.4\%$). The common-effect estimate (HR = 1.23) is shown as a sensitivity analysis.

Figure S16 | Similarity network fusion (SNF). Consensus clustering across CPTAC proteome, acetylome, and transcriptome; clusters recapitulate WEBS stratification (Kruskal?Wallis p = 7.1 × 10??).

Figure S17 | WGCNA module?WEBS association (CPTAC proteome). The most WEBS-correlated module (ME3, r = 0.72, $q = 3.7 \times 10?^{1}?$) is neural-lineage- rather than spliceosome-enriched; the spliceosome signal is distributed, not module-confined (negative control).

Figure S18 | STRING protein?protein interaction network. The spliceosome protein set forms a denser network than chance expectation (121 nodes, 3,349 edges, enrichment p < 10???; hubs SNW1, SRSF1, SF3B1).

Figure S19 | Pathway-level SHAP aggregation. SHAP values of the six-gene Cox model aggregated to Hallmark pathways; hypoxia and glycolysis programs dominate risk contribution.

Figure S20 | Genome-wide discordance tail. Distribution of per-gene protein?mRNA discordance; the spliceosome lies in the extreme tail (p = 4.7 × 10???).

Figure S21 | Pseudoreplication control. Per-patient Spearman correlations for the two key single-cell associations (WEBS vs Hallmark Hypoxia; WEBS vs Slingshot pseudotime), each point = one patient. Hypoxia decoupling is confirmed at patient level (median ? ? 0, one-sample Wilcoxon p = 1); the pseudotime trend attenuates at patient level (p = 0.22) and is presented as cell-level supportive evidence. Pseudobulk mixed-effects models (patient as random intercept) confirm the Neftel-state gradient (p = 7.4 × 10??).

Figure S22 | Per-gene RNA vs protein correlation with WEBS. Each point is one spliceosome gene (n = 109 with both layers); 107/109 show stronger protein- than mRNA-layer correlation. Bootstrap of the pathway-level ?? (2,000 resamples): 95% CI 0.60?1.03.

Figure S23 | Cross-cohort replication of WEBS-associated splicing events. Effect sizes (PSI vs WEBS Spearman ?) in TCGA-GBM vs TCGA-LGG for the 4,285 testable GBM-identified events; 83% sign-concordant (binomial $p < 10?^{1}?$), effect-size correlation ? = 0.70.

Sheet	Content
S_CPTAC_WEBS	Protein-level WEBS and group assignment for 99 CPTAC-GBM samples
S_CPTAC_limma	Full limma differential abundance results (11,141 proteins, LHS vs LLS)
S_spliceosome	Spliceosome protein subset results (111 quantified, 107 up-regulated)
S_acetyl_sites	Site-level acetylome differential analysis (8,350 sites)
S_acetyl_corr	Per-site acetylation?WEBS correlations
S_bulk_survival	WEBS survival analysis across bulk cohorts
S_signature_valid	Six-gene signature validation across cohorts (fixed coefficients)
S_bootstrap	Bootstrap selection frequencies (100 resamples)
S_SHAP	SHAP values summary per gene and risk group
S_pancancer	Pan-cancer WEBS?survival summary (4 TCGA cohorts)
S_benchmark	Method correlation matrix (WEBS/AUCell/ssGSEA/PLAGE/mean)
S_performance	Computational runtime and memory at 1k?35.8k cells
S_sensitivity_maxrank	UCell maxRank sweep
S_sensitivity_loo	Leave-one/two-out robustness (57 perturbations)
S_specificity	Spliceosome vs random-pathway discordance control
S_spatial	Moran's I / Geary's c spatial autocorrelation per slice
S_MOFA2	MOFA2 factor?WEBS correlations (4-view CPTAC integration)
S_DepMap	WEBS?dependency correlations for candidate targets
S_GDSC2_drugs	WEBS?sensitivity correlations across 286 drugs
S_GDSC2_HDAC	HDAC-inhibitor subset results
S_combinations	Bliss-independence combination predictions
S_CMap	Drugs targeting LHS-up proteins with WEBS-linked sensitivity
S_CellChat	Pathway-level information flow LLS vs LHS
S_MIF_perm	MIF?CD74/CXCR4 permutation test
S_docking	Molecular docking binding energies (4 target?ligand pairs)
S_ablation	Ablation study: leave-one-out and single-gene C-indices of the six-gene signature (TCGA-GBM)
S_splicing_sig4387	Complete list of the 4,387 WEBS-associated alternative splicing events (event ID, gene, type, rho, p, q)
S_splicing_top	Top 500 WEBS-associated splicing events by |rho| (subset of S_splicing_sig4387)
S_pancancer33	Pan-cancer WEBS?spliceosome correlation and signature HR across 33 TCGA cohorts
S_virtualKO	GENIE3 connectivity sensitivity summary (absolute shift ratios per regulator)
S_glass_longitudinal	GLASS primary?recurrence paired WEBS data (179 pairs, 148 patients)
S_glass_primary	GLASS primary-tumor WEBS and risk scores with survival (n = 107)
S_phospho_spliceosome	Spliceosome phosphosite statistics (raw and abundance-corrected)
S_splicing_enrichment	GO enrichment of genes harboring WEBS-associated splicing events

| S_Table2_targets | Candidate therapeutic vulnerability prioritization (Table 2) |

| S_glass_design | GLASS analysis units: sample sizes, units, and inclusion criteria per analysis |

| S_signature_performance | Six-gene signature performance across all cohorts: per-SD HR (95% CI), C-index, time-dependent AUC (12/36/60 months). Note: per-SD HRs are standardized within each cohort; the per-unit HRs quoted in the main text for GSE7696/GSE83300 use the raw risk-score scale and are therefore not directly comparable in magnitude |

| S_threshold_sensitivity | Spliceosome up-regulation robustness across WEBS stratification thresholds (109?111/111 up-regulated) |

| S_protein_WEBS | Protein-level WEBS composition: 10 writer and 16 eraser proteins quantified in the CPTAC TMT panel |

- snATAC-seq: fragment files were not deposited for GSE138794; accessibility analysis is therefore limited to peak-matrix promoter aggregation (TSS ± 2 kb), and nucleosome/TSS quality metrics could not be computed.
- RNA velocity: spliced/unspliced matrices are not deposited for GSE138794; trajectory evidence relies on Slingshot pseudotime.
- REMBRANDT / GSE13041: these cohorts lack survival fields in their GEO series matrices and were omitted without affecting central claims. GLASS data were accessed via Synapse (syn17038081) under the consortium's data-use terms and are included for validation and longitudinal analyses (main text Figs. 13?14).
- RCTD deconvolution: spatial deconvolution was performed with Seurat label transfer; RCTD was attempted but the package failed at its internal caching step on Windows (choose_sigma_c); results were concordant with label-transfer deconvolution.

Figure S24 | External single-cell validation (GSE131928). WEBS by Neftel state in Smart-seq2 and 10x platforms (KW $p = 1.6 \times 10^{?1??}$ and $1.0 \times 10^{?313}$), cross-platform patient consistency (? = 0.67), and LASSO cross-validation/coefficient paths for the six-gene signature.

Figure S25 | WEBS by Neftel malignant state. Module scores by WEBS group in 7,952 neoplastic cells; WEBS vs mesenchymal module (? = 0.22).

Figure S26 | WEBS vs hypoxia and tumor regions. WEBS across region types; patient-level paired Margin vs Core; WEBS vs CA9/VEGFA; Hallmark Hypoxia by WEBS group.

Figure S27 | Pseudotime trajectory. Slingshot on neoplastic cells (cell-level association; attenuated at patient level, see Fig. S21).

Figure S28 | Cell?cell communication across WEBS states. CellChat comparison (SPP1?CD44 up in high-WEBS; MIF?CD74/CXCR4 enriched in low-WEBS).

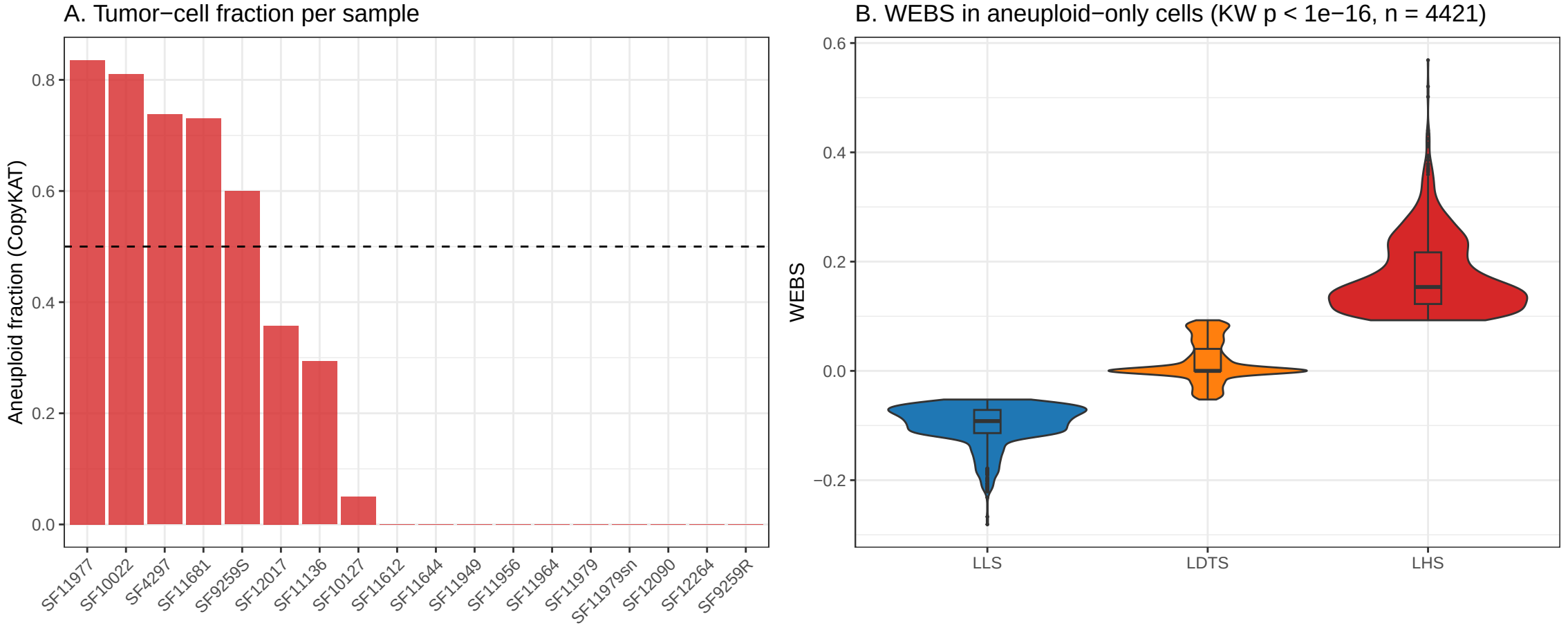
Figure S29 | Spatial transcriptomics. Per-spot WEBS on six Visium GBM slices; significant autocorrelation in 3/6 slices.

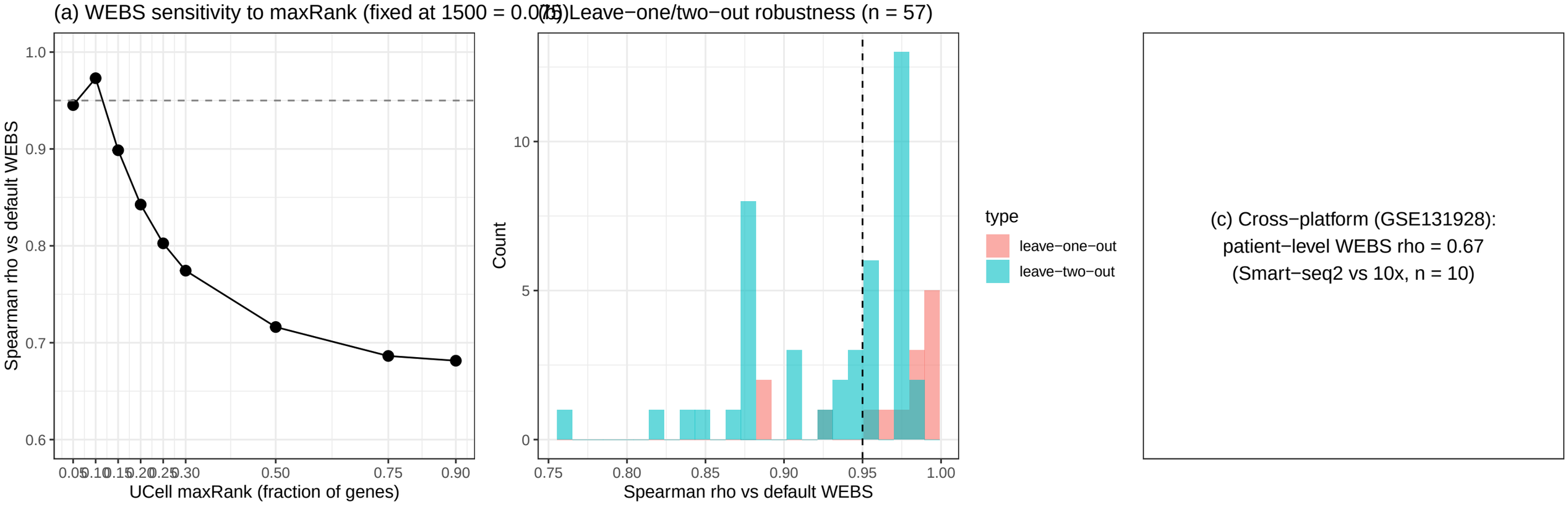
Figure S30 | snATAC-seq promoter accessibility. Writer/eraser promoter accessibility vs patient-median WEBS (no association; peak-matrix level).

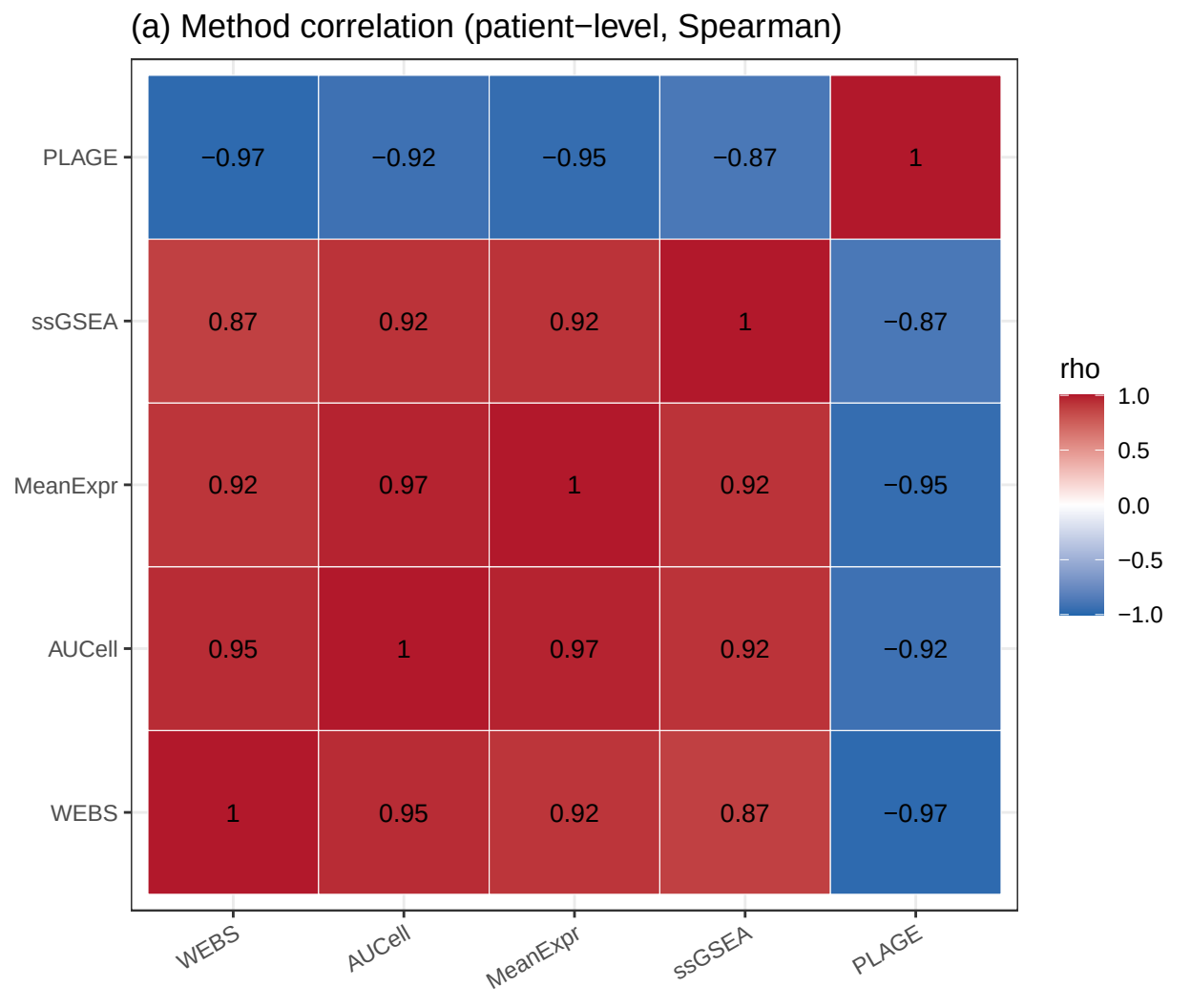
Figure S31 | Functional characterization of WEBS states. Hallmark pathway activity, CytoTRACE-like stemness, perturbation sensitivity, differential TFs.

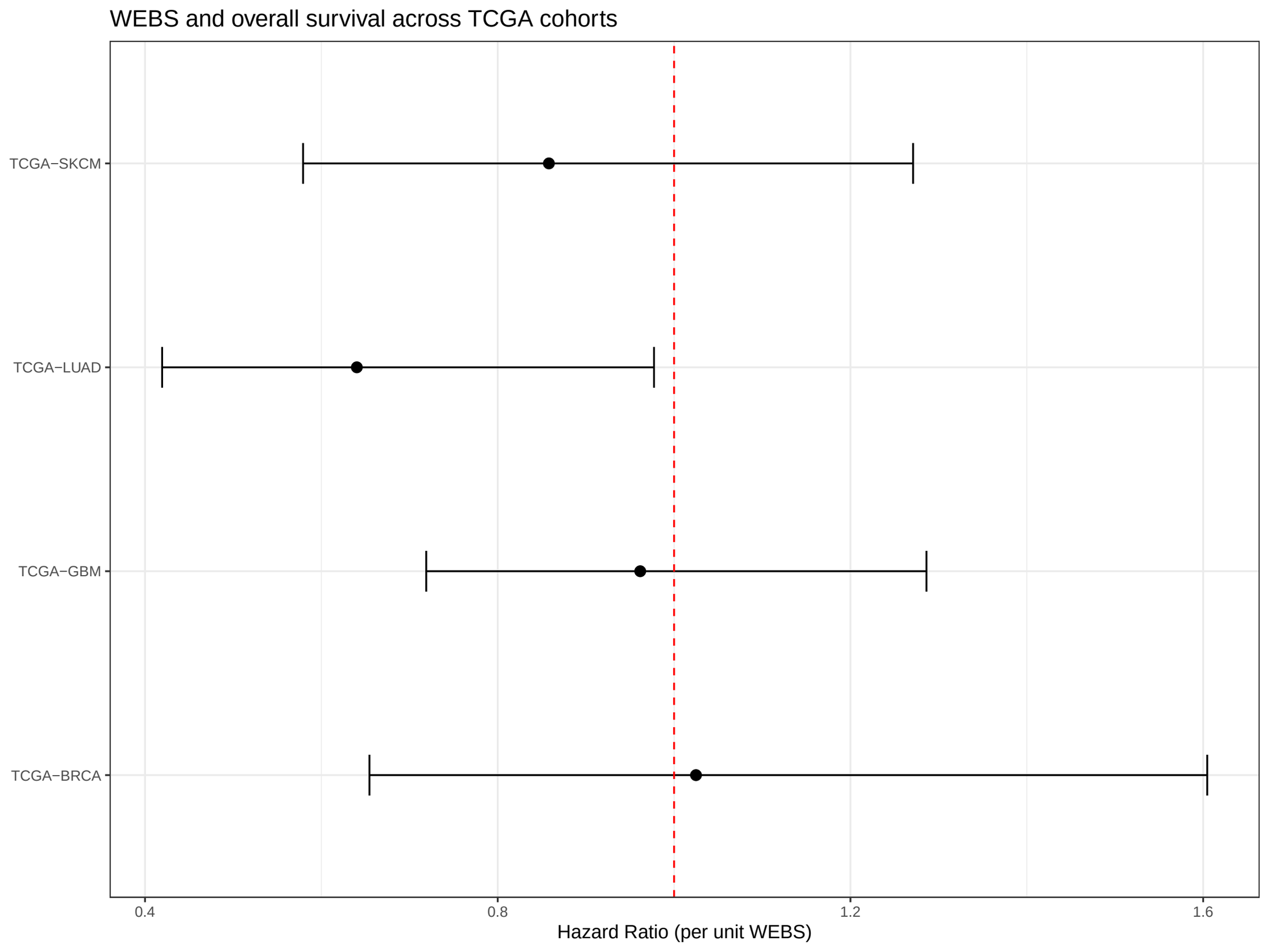
Figure S32 | Pan-cancer scope. WEBS?spliceosome correlation and six-gene signature HR across 33 TCGA cohorts (GBM highlighted).

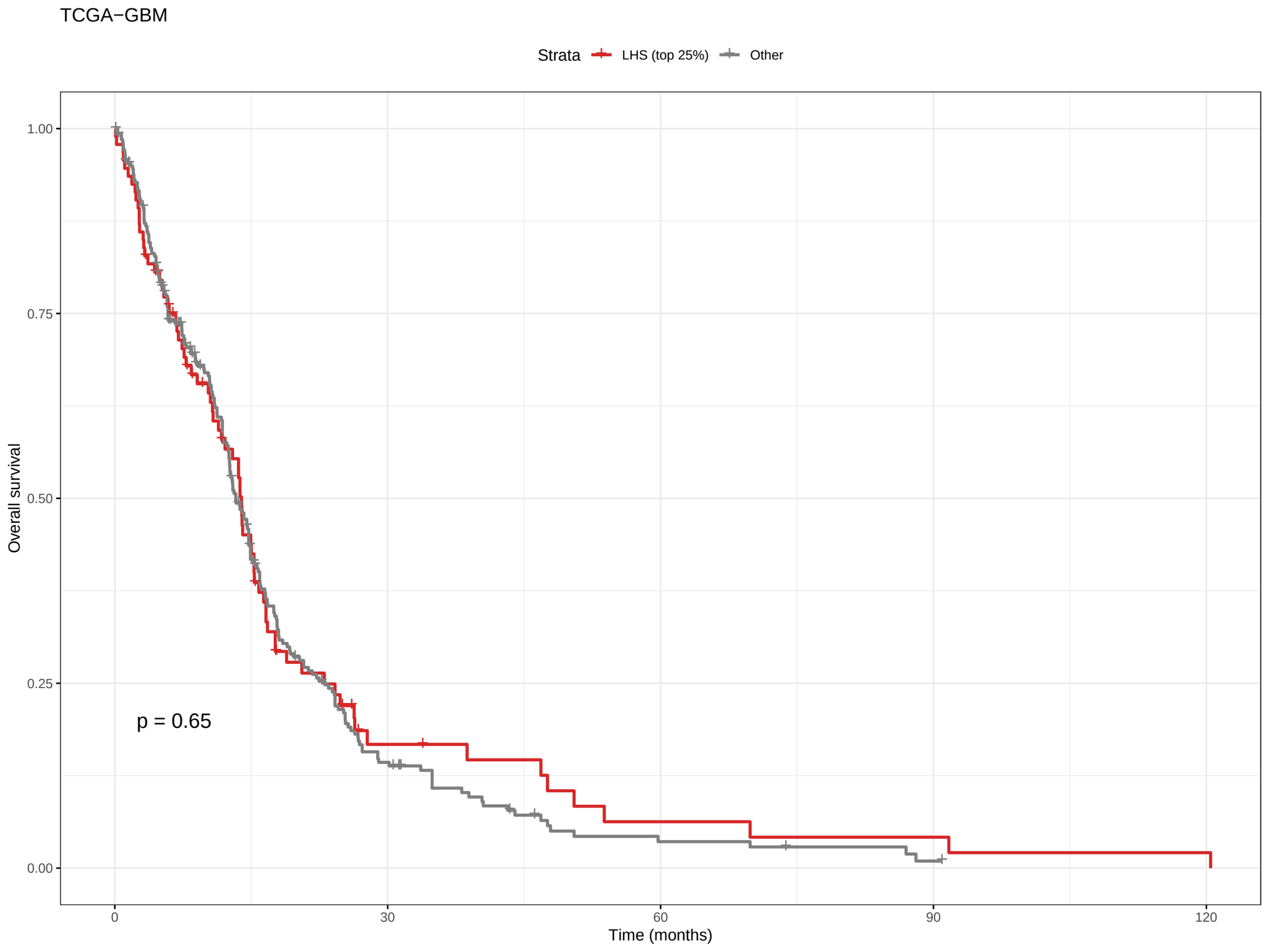
Figure S33 | Working model. Writer?eraser balance axis and proposed post-transcriptional spliceosome regulation in GBM.

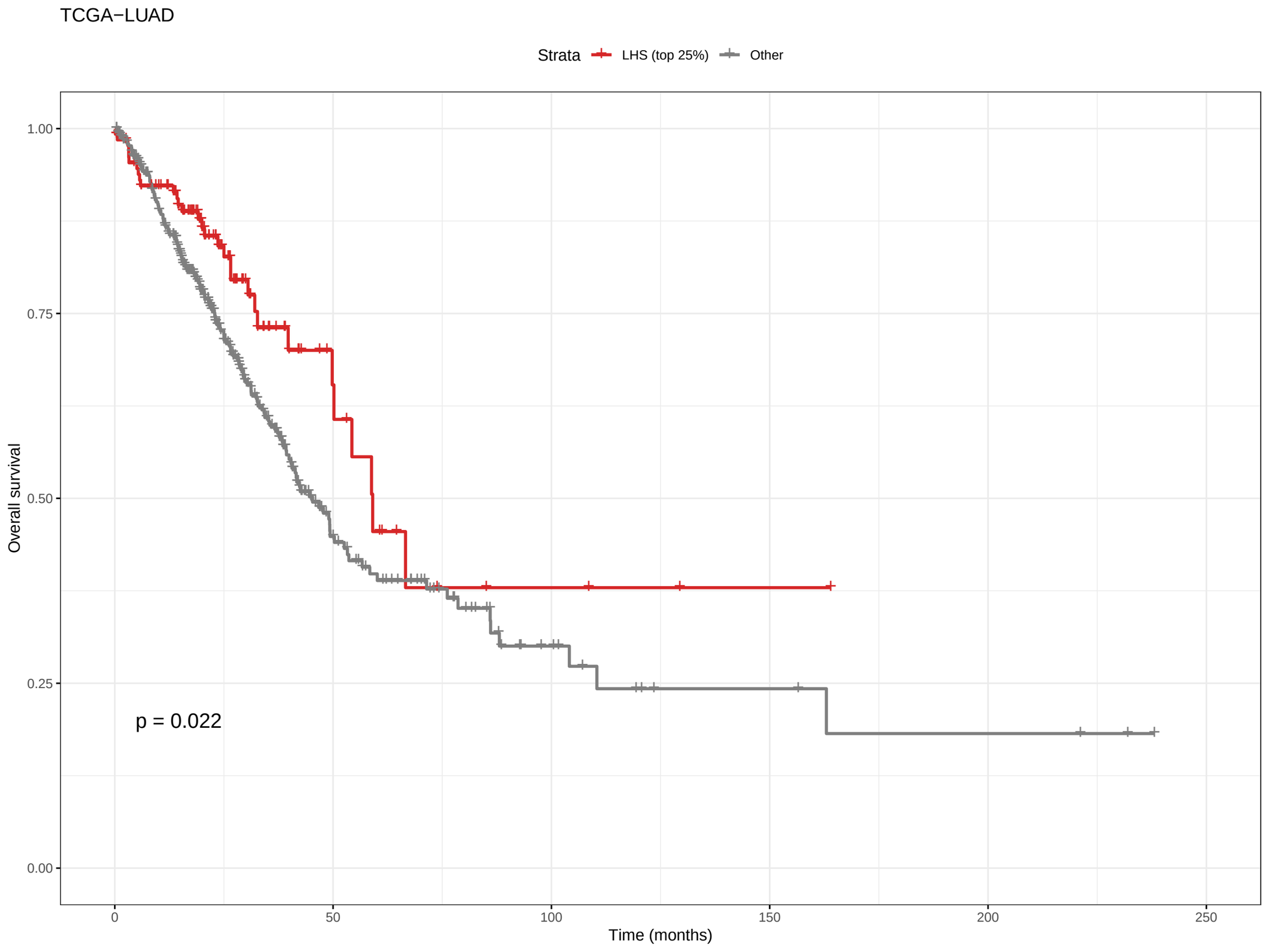

A. Tumor−cell fraction per sample
Aneuploid fraction (CopyKAT)
0.0
0.2
0.4
0.6
0.8
SF11977
SF10022
SF4297
SF11681
SF9259S
SF12017
SF11136
SF10127
SF11612
SF11644
SF11949
SF11956
SF11964
SF11979
SF11979sn
SF12090
SF12264
SF9259R
B. WEBS in aneuploid−only cells (KW p < 1e−16, n = 4421)
WEBS
−0.2
0.0
0.2
0.4
0.6
LLS
LDTS
LHS

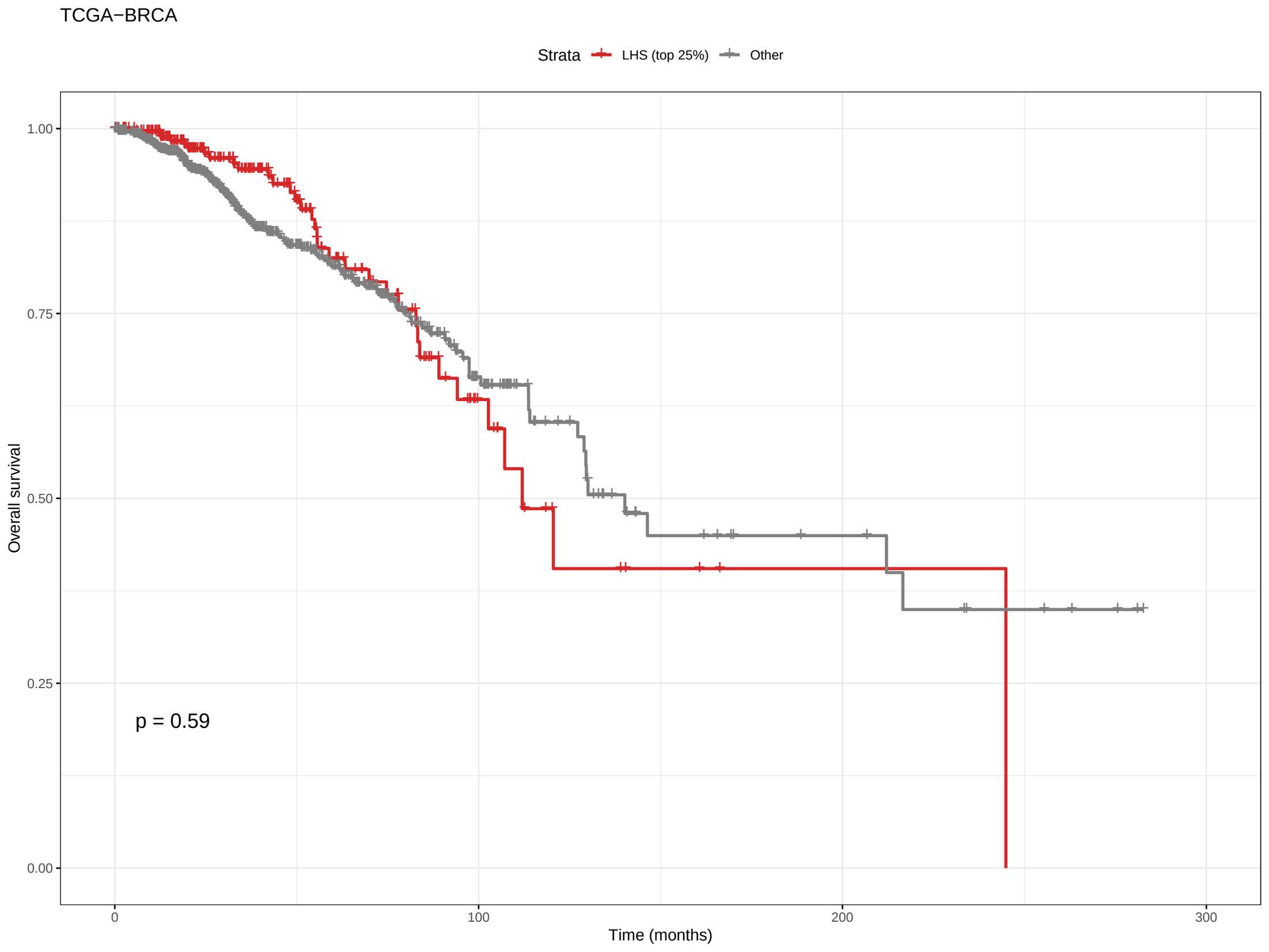

(a) WEBS sensitivity to maxRank (fixed at 1500 = 0.075)
Spearman rho vs default WEBS
1.0
0.9
0.8
0.7
0.6
0.05
0.10
0.15
0.20
0.25
0.30
0.50
0.75
0.90
UCell maxRank (fraction of genes)
(b) Leave−one/two−out robustness (n = 57)
Count
10
5
0
0.75
0.80
0.85
0.90
0.95
1.00
Spearman rho vs default WEBS
type
leave−one−out
leave−two−out
(c) Cross−platform (GSE131928):
patient−level WEBS rho = 0.67
(Smart−seq2 vs 10x, n = 10)

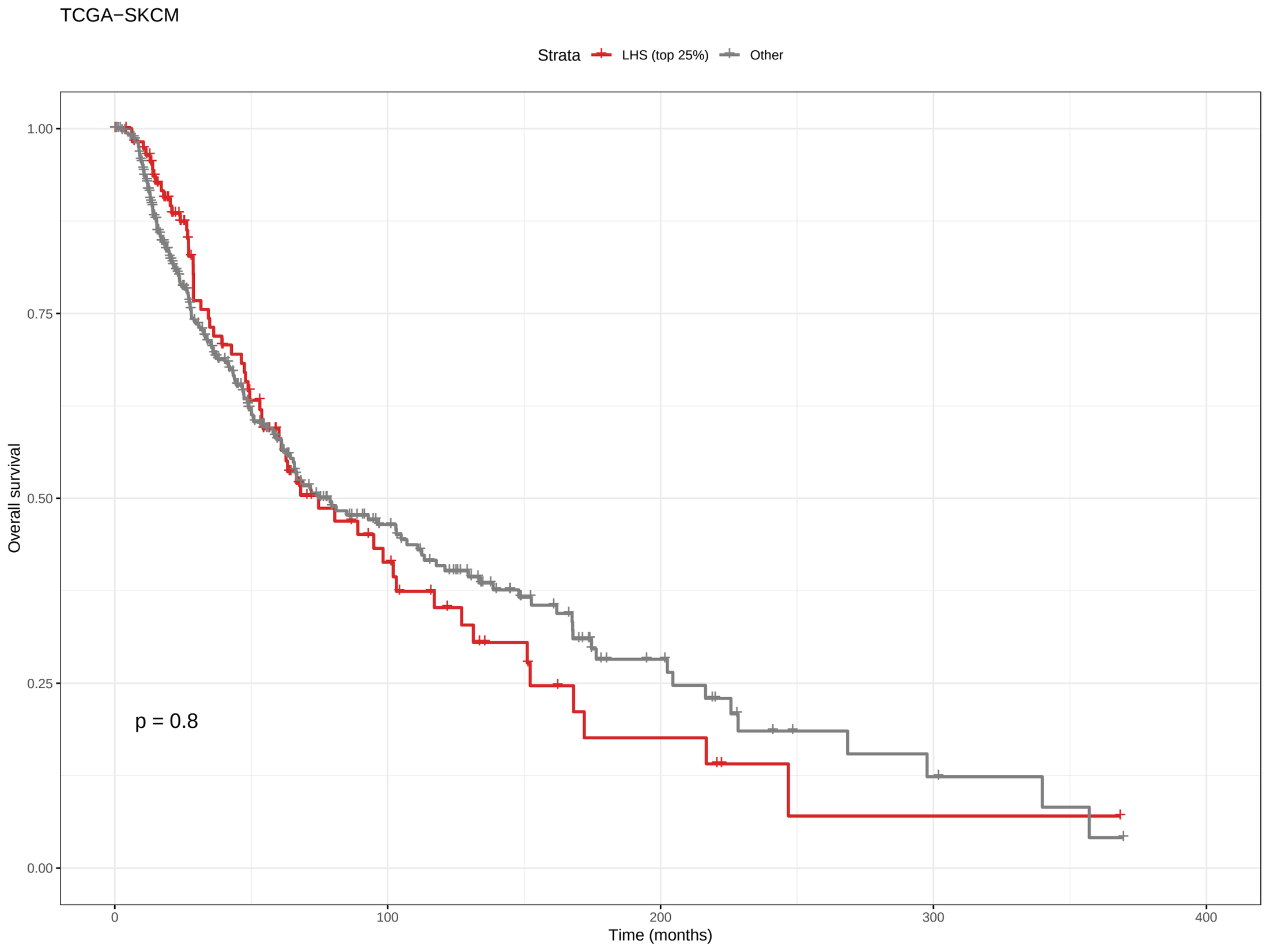

(a) Method correlation (patient−level, Spearman)
PLAGE
ssGSEA
MeanExpr
AUCell
WEBS
−0.97
−0.92
−0.95
−0.87
1
0.87
0.92
0.92
1
−0.87
0.92
0.97
1
0.92
−0.95
0.95
1
0.97
0.92
−0.92
1
0.95
0.92
0.87
−0.97
WEBS
AUCell
MeanExpr
ssGSEA
PLAGE
rho
1.0
0.5
0.0
−0.5
−1.0

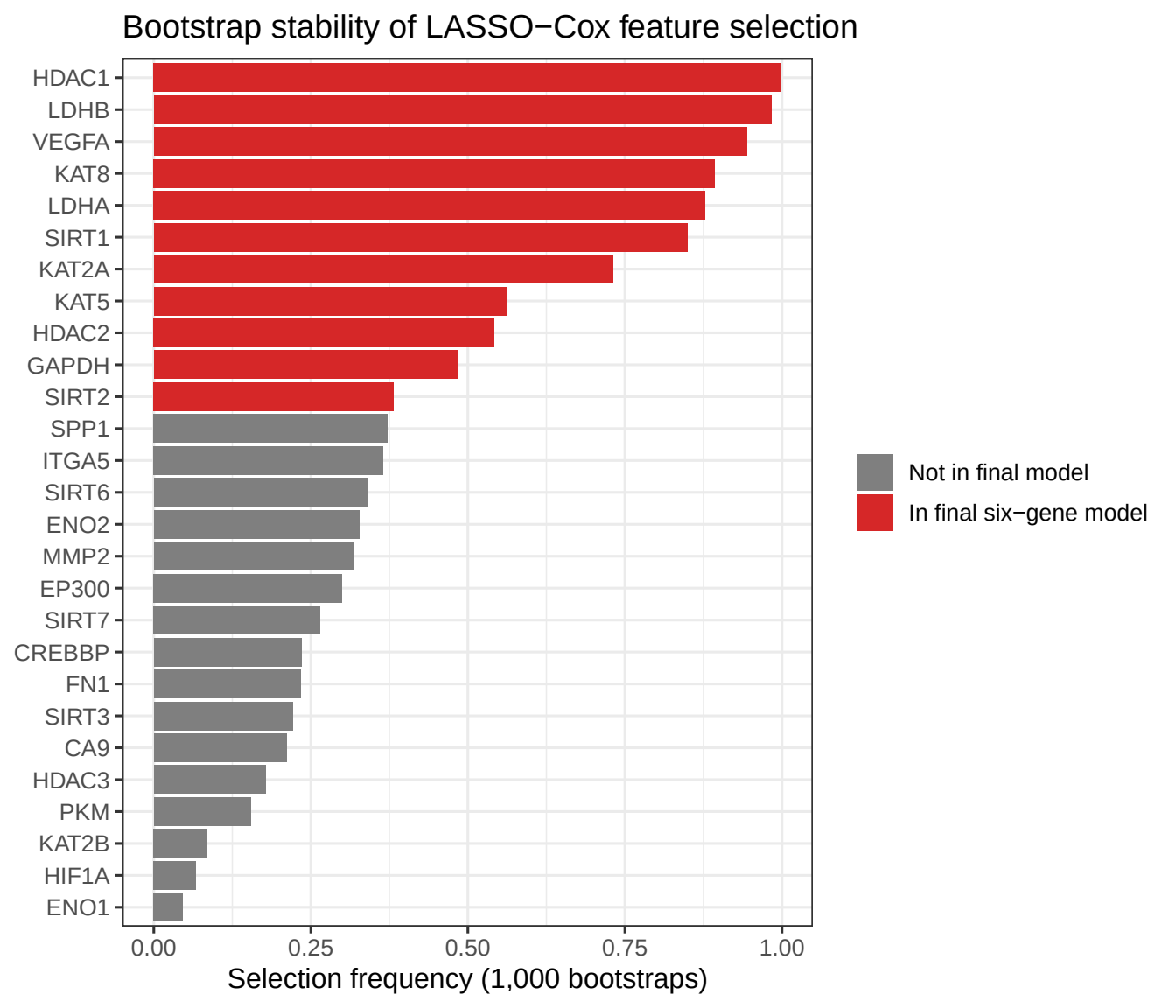

WEBS and overall survival across TCGA cohorts
TCGA−SKCM
TCGA−LUAD
TCGA−GBM
TCGA−BRCA
0.4
0.8
1.2
1.6
Hazard Ratio (per unit WEBS)

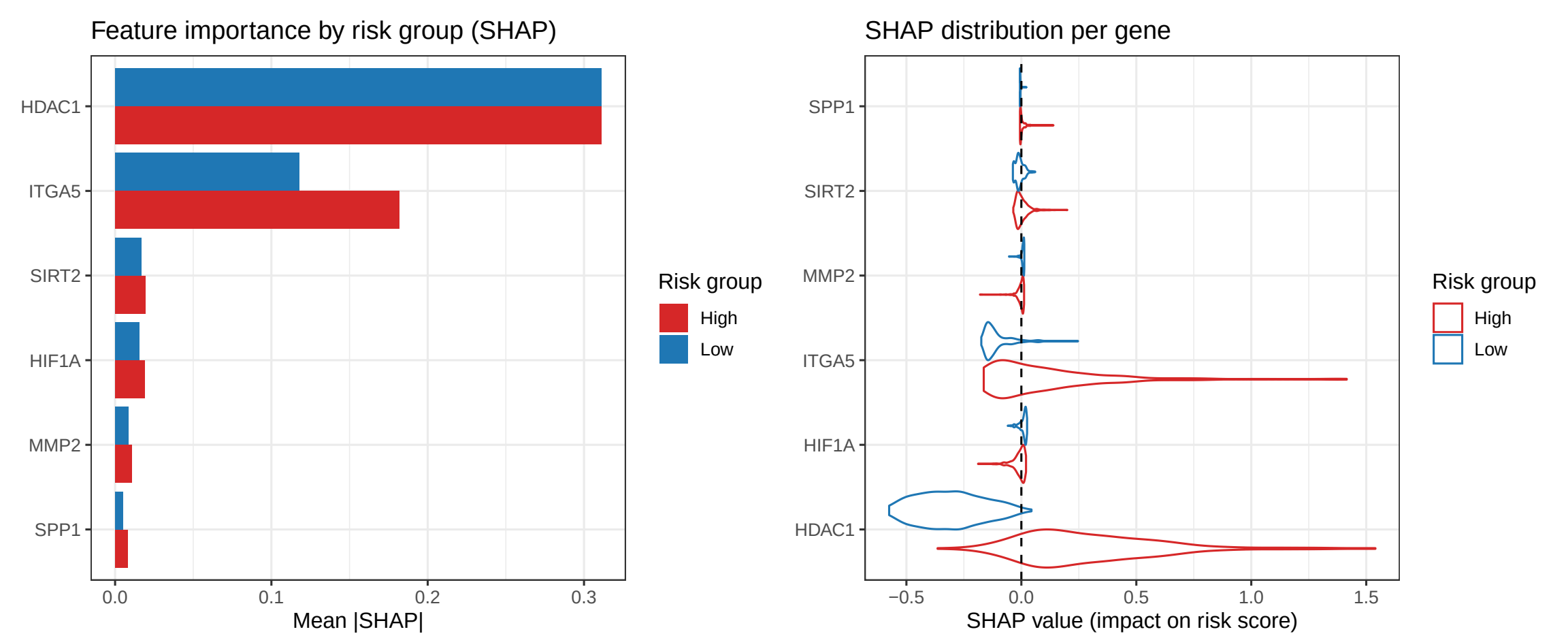

TCGA−GBM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.65
0
30
60
90
120
Time (months)

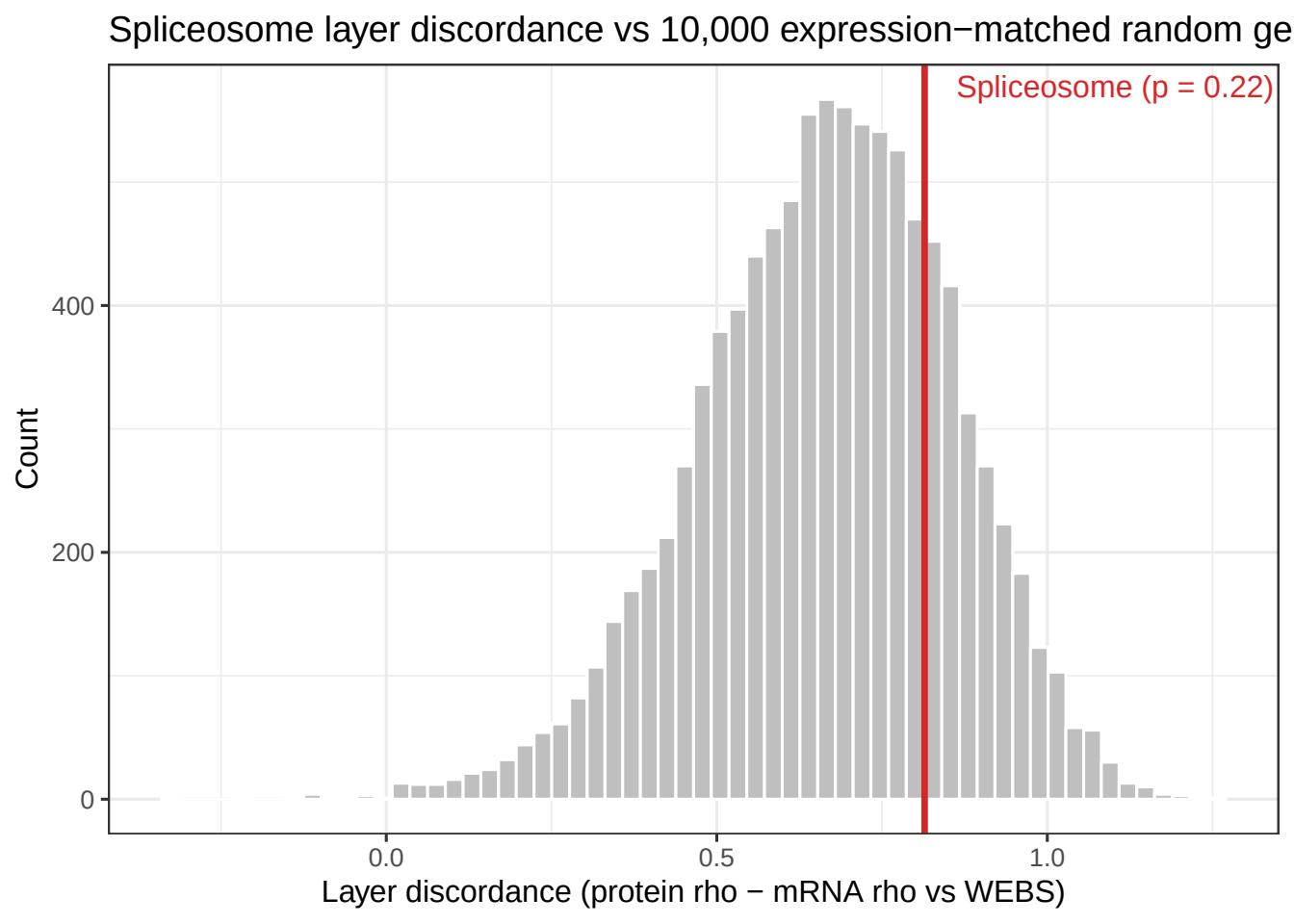

TCGA−LUAD
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.022
0
50
100
150
200
250
Time (months)

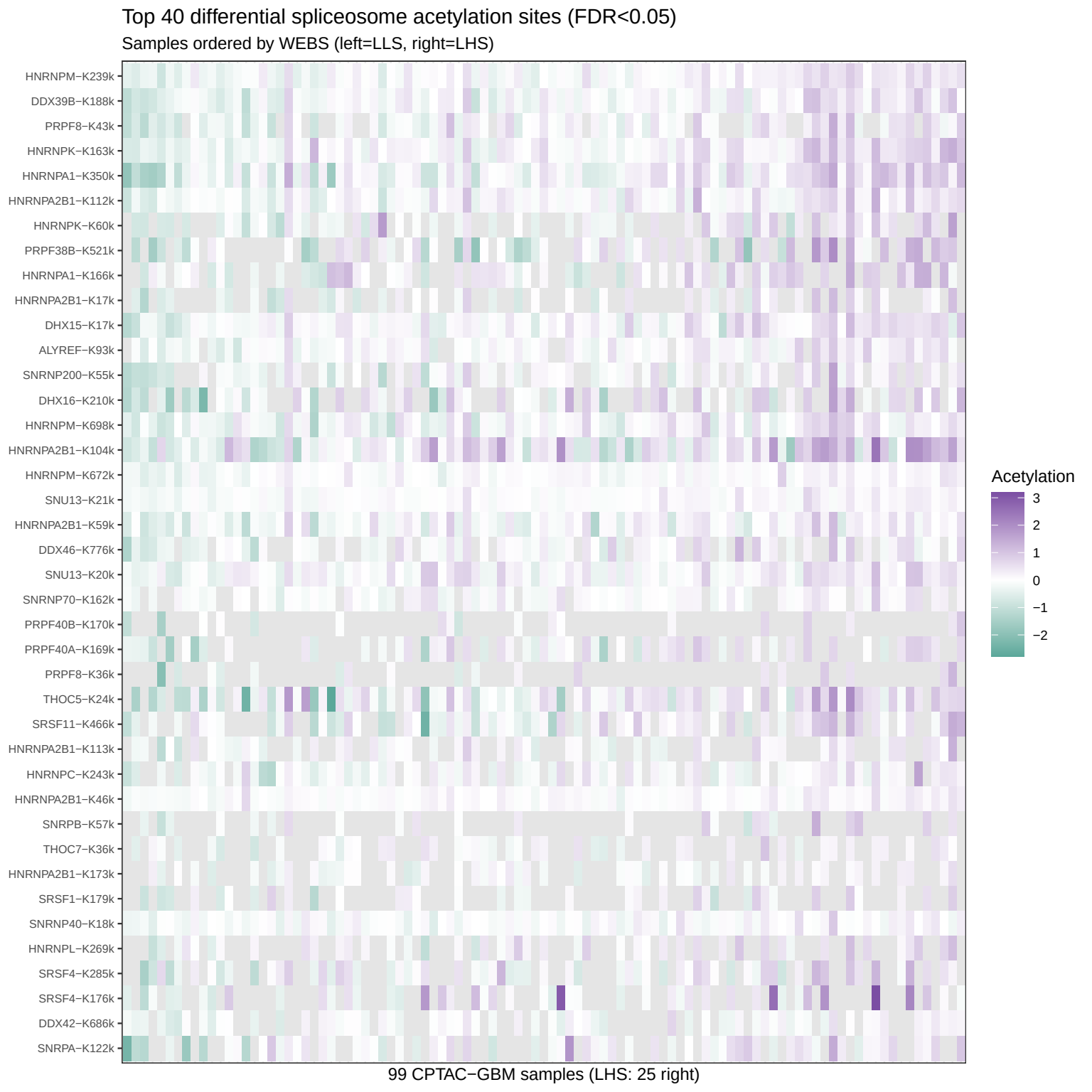

TCGA−BRCA
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.59
0
100
200
300
Time (months)

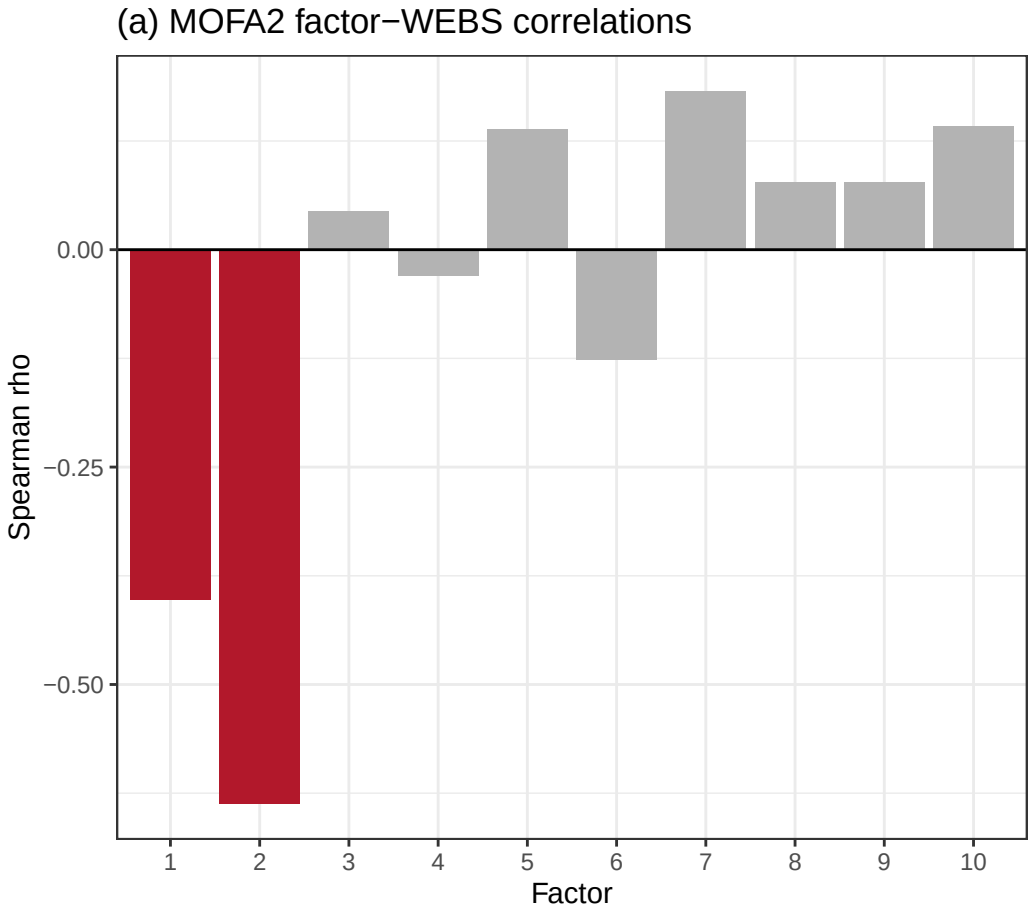

TCGA−SKCM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.8
0
100
200
300
400
Time (months)

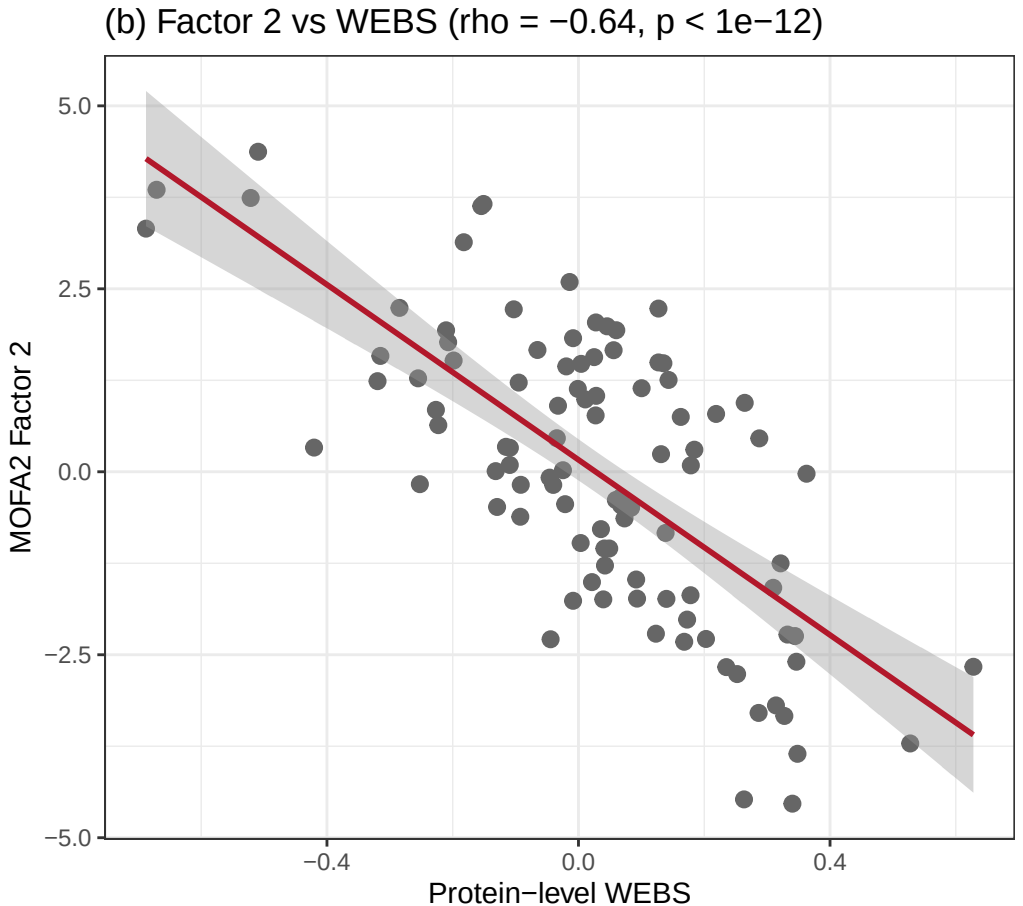

Bootstrap stability of LASSO−Cox feature selection
HDAC1
LDHB
VEGFA
KAT8
LDHA
SIRT1
KAT2A
KAT5
HDAC2
GAPDH
SIRT2
SPP1
ITGA5
SIRT6
ENO2
MMP2
EP300
SIRT7
CREBBP
FN1
SIRT3
CA9
HDAC3
PKM
KAT2B
HIF1A
ENO1
0.00
0.25
0.50
0.75
1.00
Selection frequency (1,000 bootstraps)
Not in final model
In final six−gene model

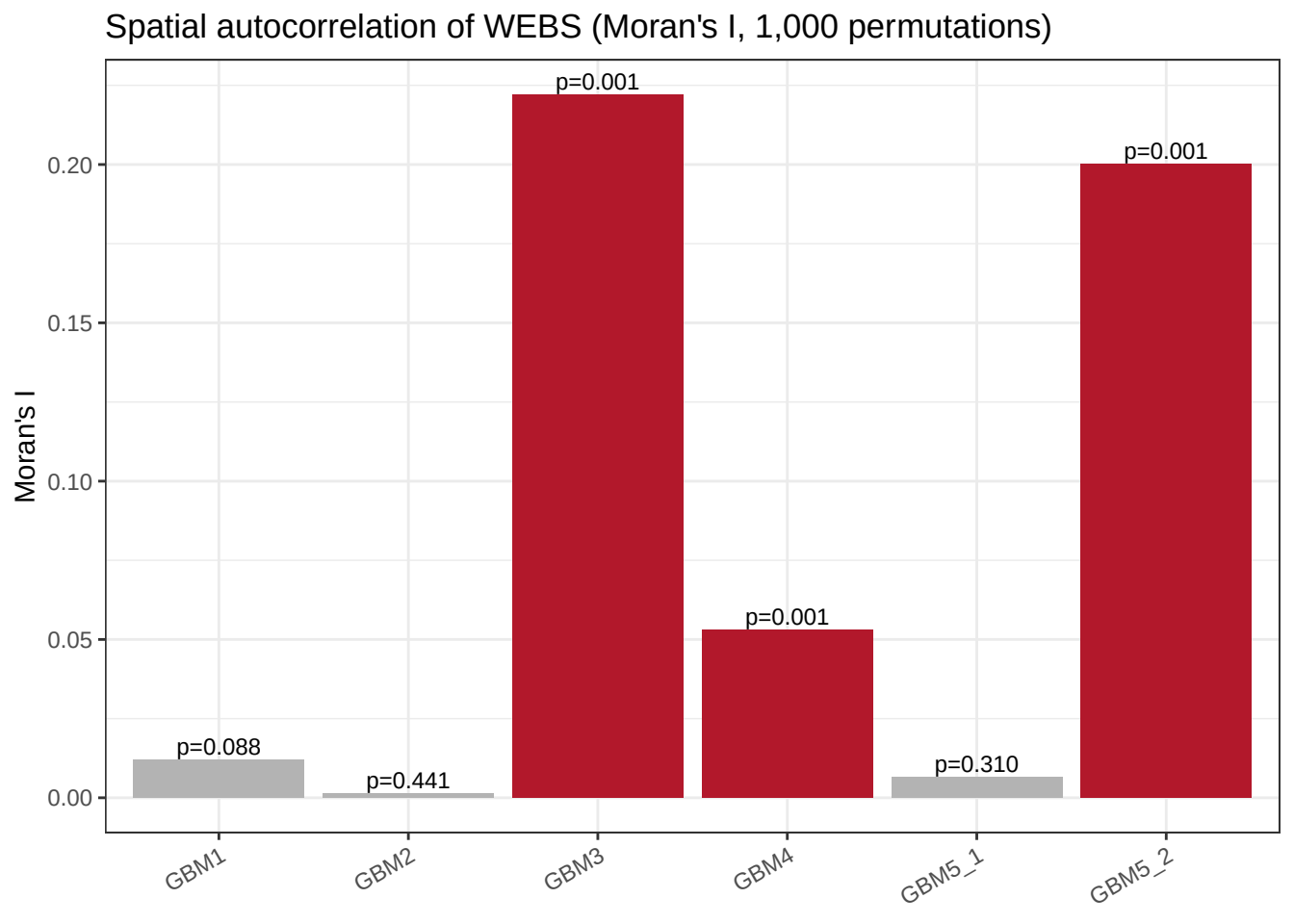

Feature importance by risk group (SHAP)
HDAC1
ITGA5
SIRT2
HIF1A
MMP2
SPP1
0.0
0.1
0.2
0.3
Mean |SHAP|
Risk group
High
Low
SHAP distribution per gene
SPP1
SIRT2
MMP2
ITGA5
HIF1A
HDAC1
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on risk score)
Risk group
High
Low

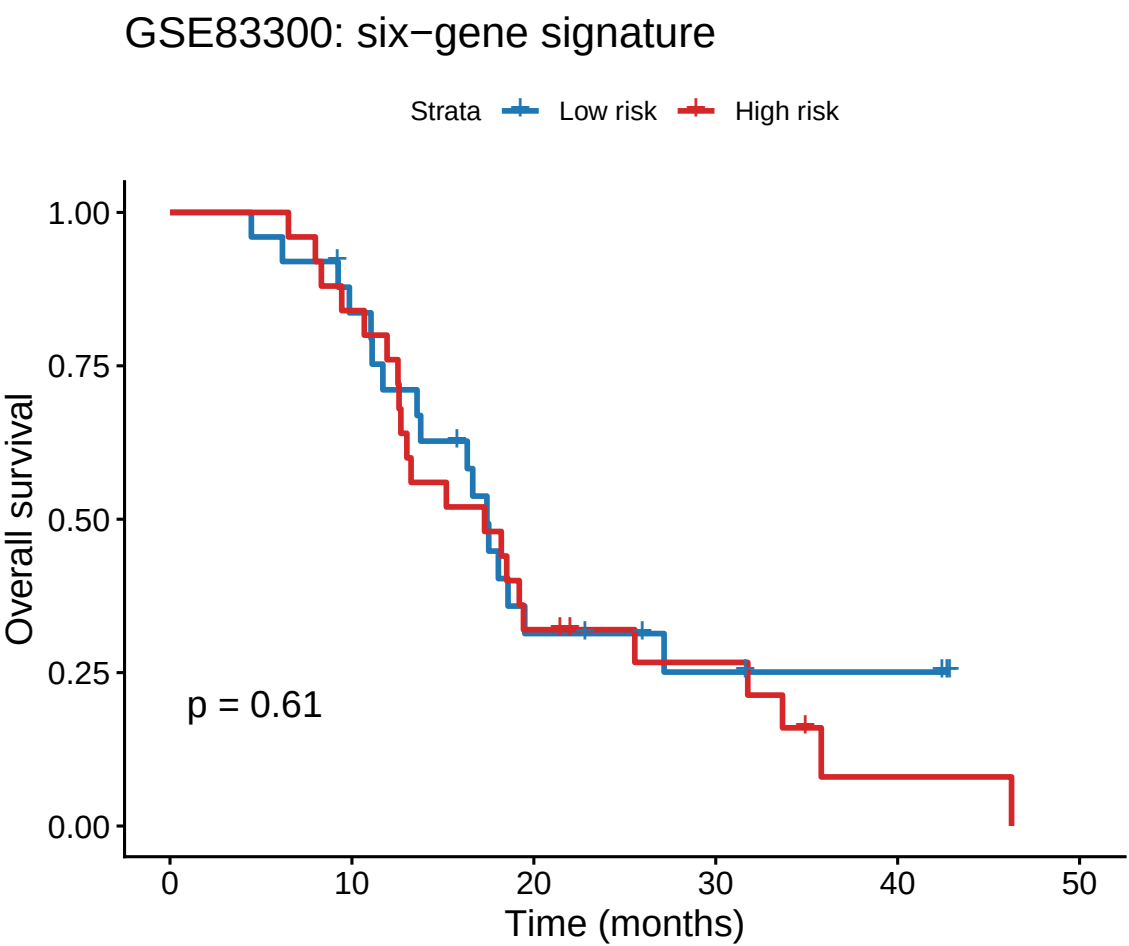

Spliceosome layer discordance vs 10,000 expression−matched random ge
Spliceosome (p = 0.22)
Count
400
200
0
0.0
0.5
1.0
Layer discordance (protein rho − mRNA rho vs WEBS)

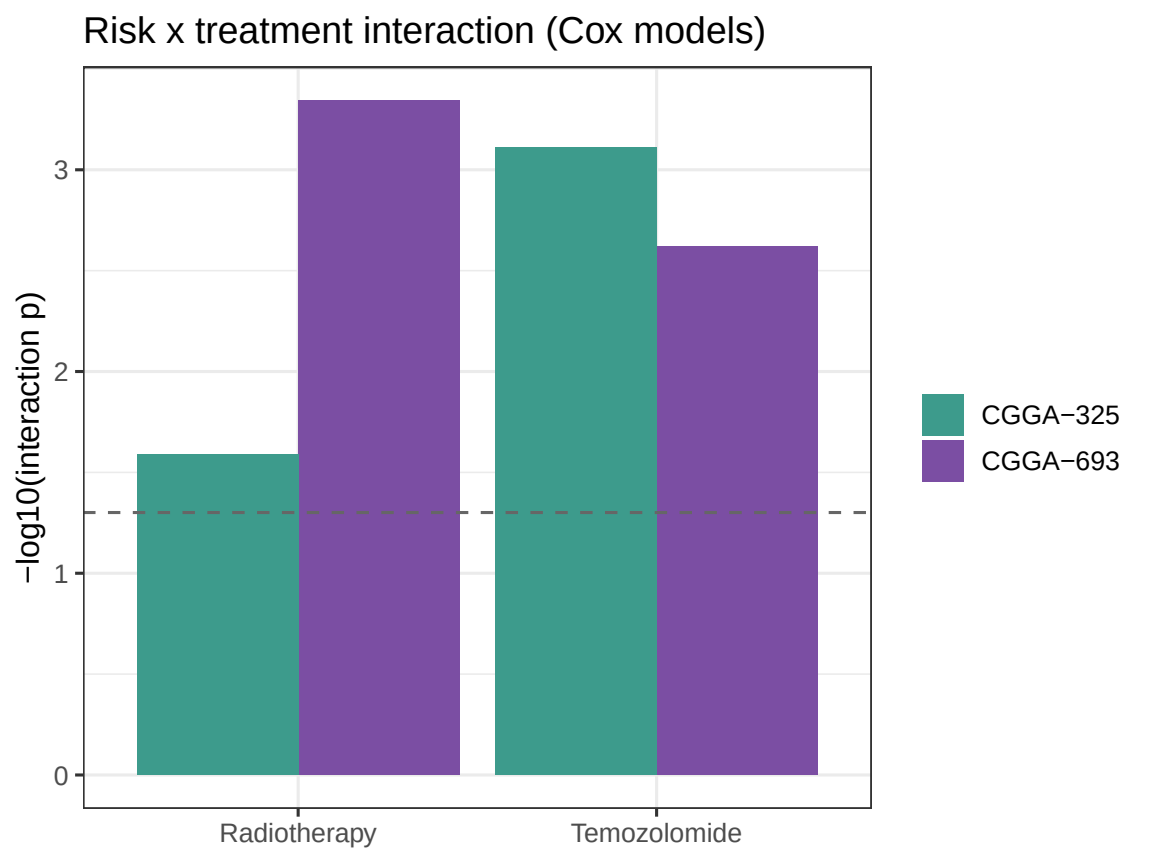

Top 40 differential spliceosome acetylation sites (FDR<0.05)
Samples ordered by WEBS (left=LLS, right=LHS)
HNRNPM−K239k
DDX39B−K188k
PRPF8−K43k
HNRNPK−K163k
HNRNPA1−K350k
HNRNPA2B1−K112k
HNRNPK−K60k
PRPF38B−K521k
HNRNPA1−K166k
HNRNPA2B1−K17k
DHX15−K17k
ALYREF−K93k
SNRNP200−K55k
DHX16−K210k
HNRNPM−K698k
HNRNPA2B1−K104k
HNRNPM−K672k
SNU13−K21k
HNRNPA2B1−K59k
DDX46−K776k
SNU13−K20k
SNRNP70−K162k
PRPF40B−K170k
PRPF40A−K169k
PRPF8−K36k
THOC5−K24k
SRSF11−K466k
HNRNPA2B1−K113k
HNRNPC−K243k
HNRNPA2B1−K46k
SNRPB−K57k
THOC7−K36k
HNRNPA2B1−K173k
SRSF1−K179k
SNRNP40−K18k
HNRNPL−K269k
SRSF4−K285k
SRSF4−K176k
DDX42−K686k
SNRPA−K122k
99 CPTAC−GBM samples (LHS: 25 right)
Acetylation
3
2
1
0
−1
−2

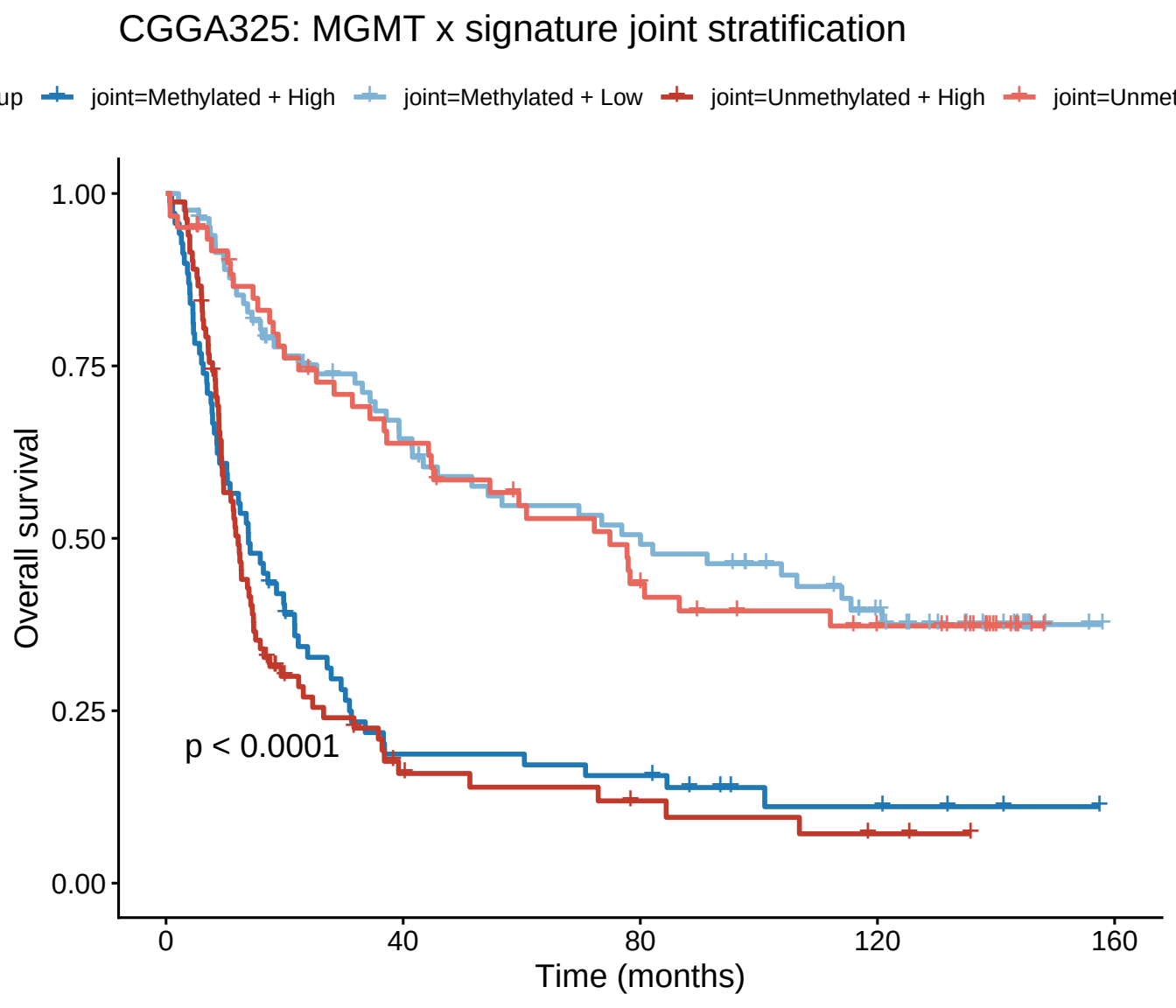

(a) MOFA2 factor−WEBS correlations
Spearman rho
0.00
−0.25
−0.50
1
2
3
4
5
6
7
8
9
10
Factor

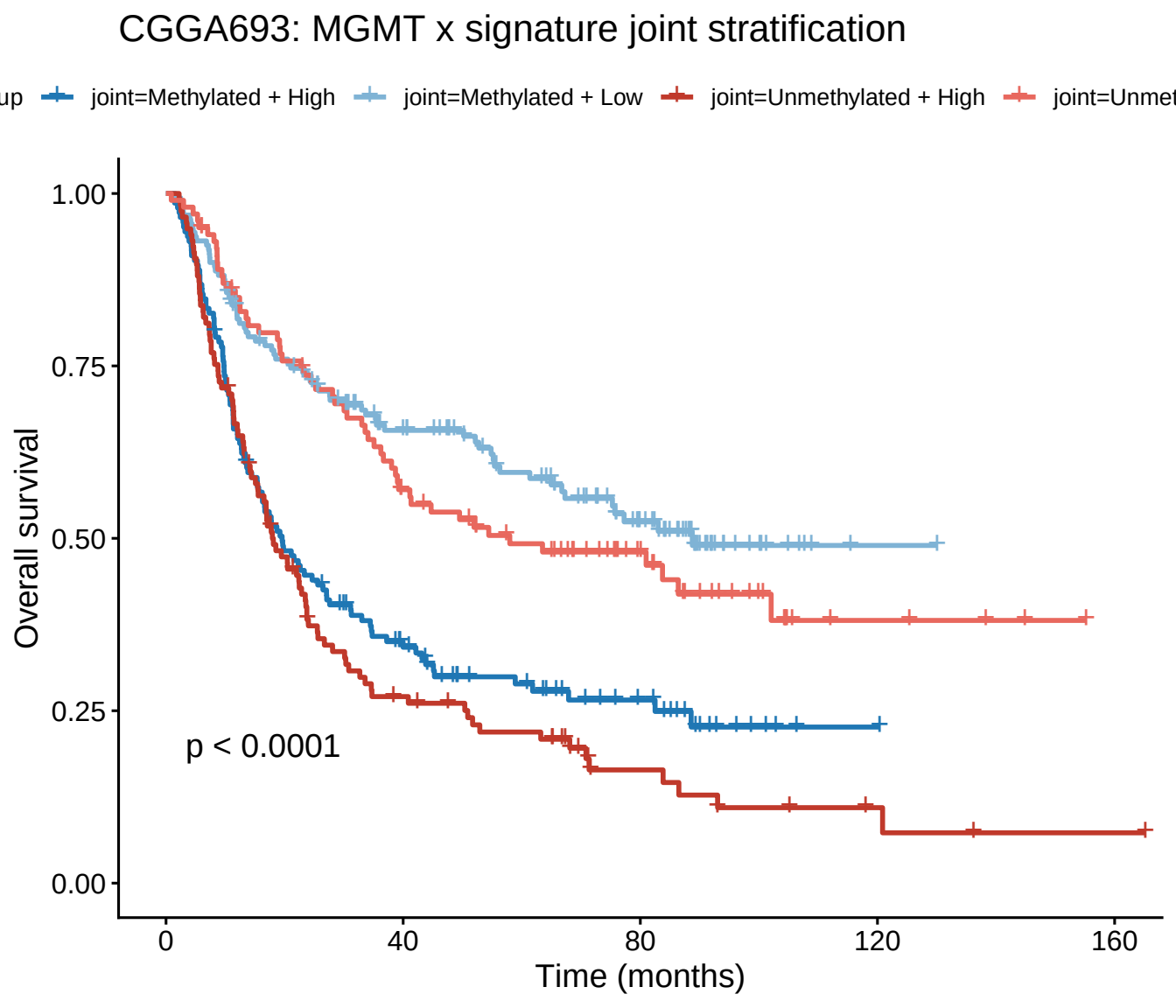

(b) Factor 2 vs WEBS (rho = −0.64, p < 1e−12)
MOFA2 Factor 2
5.0
2.5
0.0
−2.5
−5.0
−0.4
0.0
0.4
Protein−level WEBS

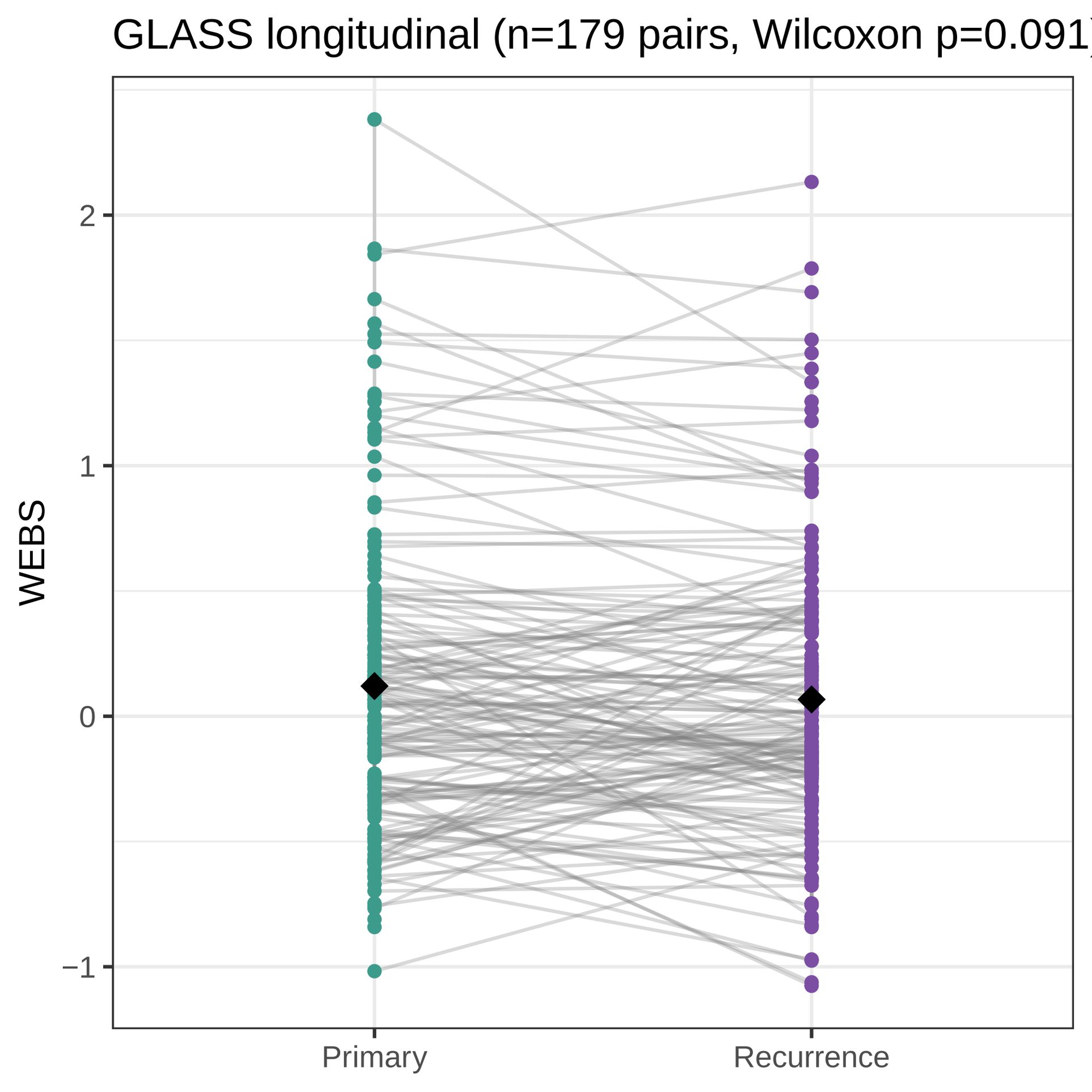

Spatial autocorrelation of WEBS (Moran's I, 1,000 permutations)
Moran's I
0.20
0.15
0.10
0.05
0.00
p=0.088
p=0.441
p=0.001
p=0.001
p=0.310
p=0.001
GBM1
GBM2
GBM3
GBM4
GBM5_1
GBM5_2

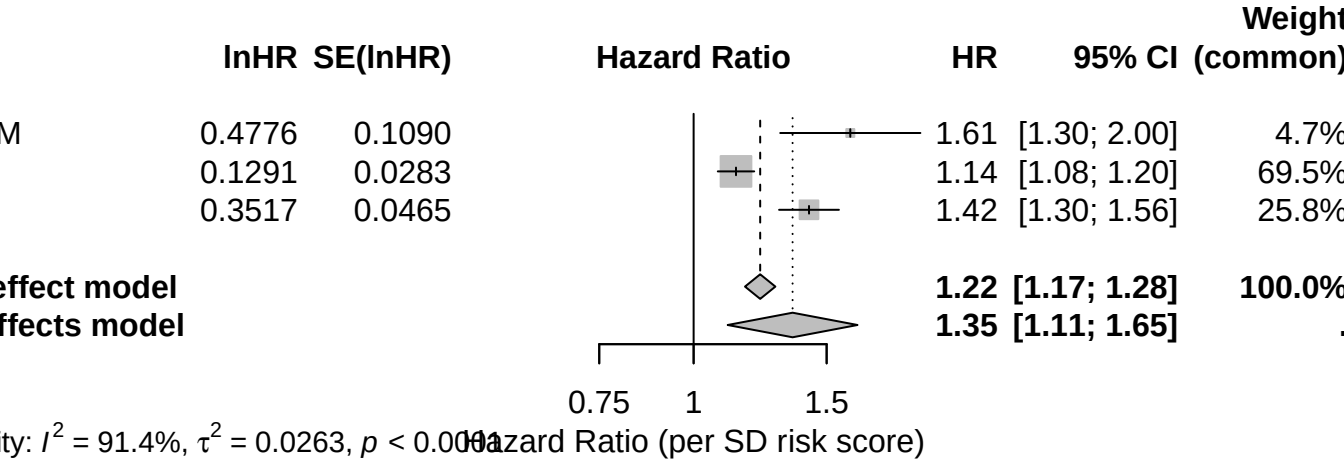

GSE83300: six−gene signature
Strata
Low risk
High risk
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.61
0
10
20
30
40
50
Time (months)

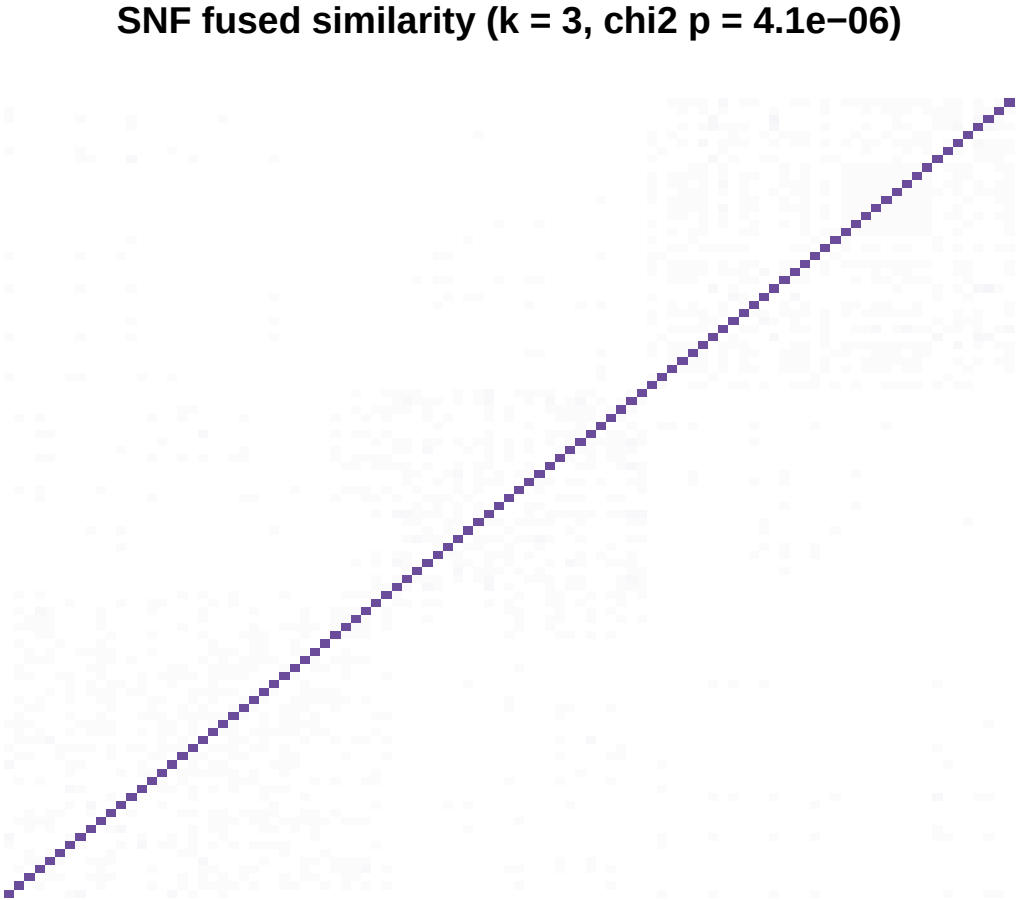

Risk x treatment interaction (Cox models)
−log10(interaction p)
0
1
2
3
Radiotherapy
Temozolomide
CGGA−325
CGGA−693

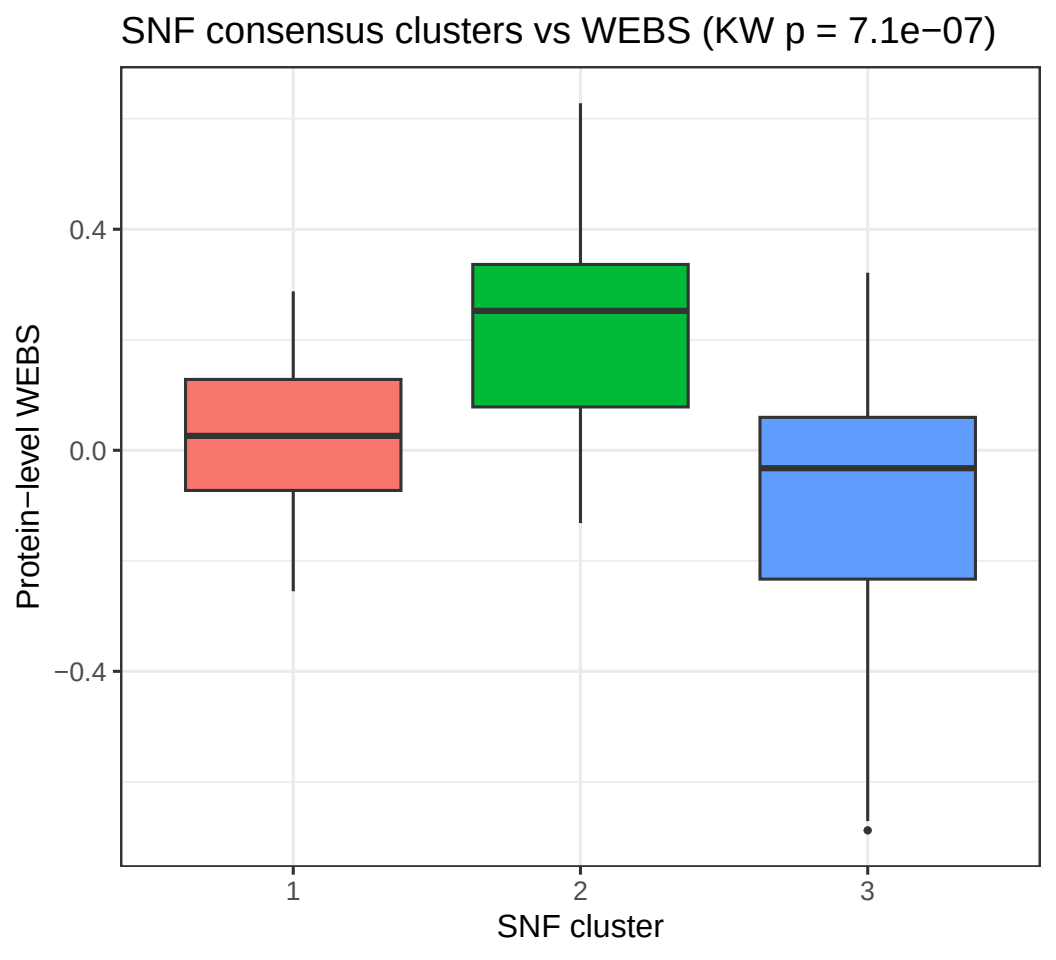

CGGA325: MGMT x signature joint stratification
up
joint=Methylated + High
joint=Methylated + Low
joint=Unmethylated + High
joint=Unmet
1.00
0.75
0.50
0.25
0.00
Overall survival
p < 0.0001
0
40
80
120
160
Time (months)

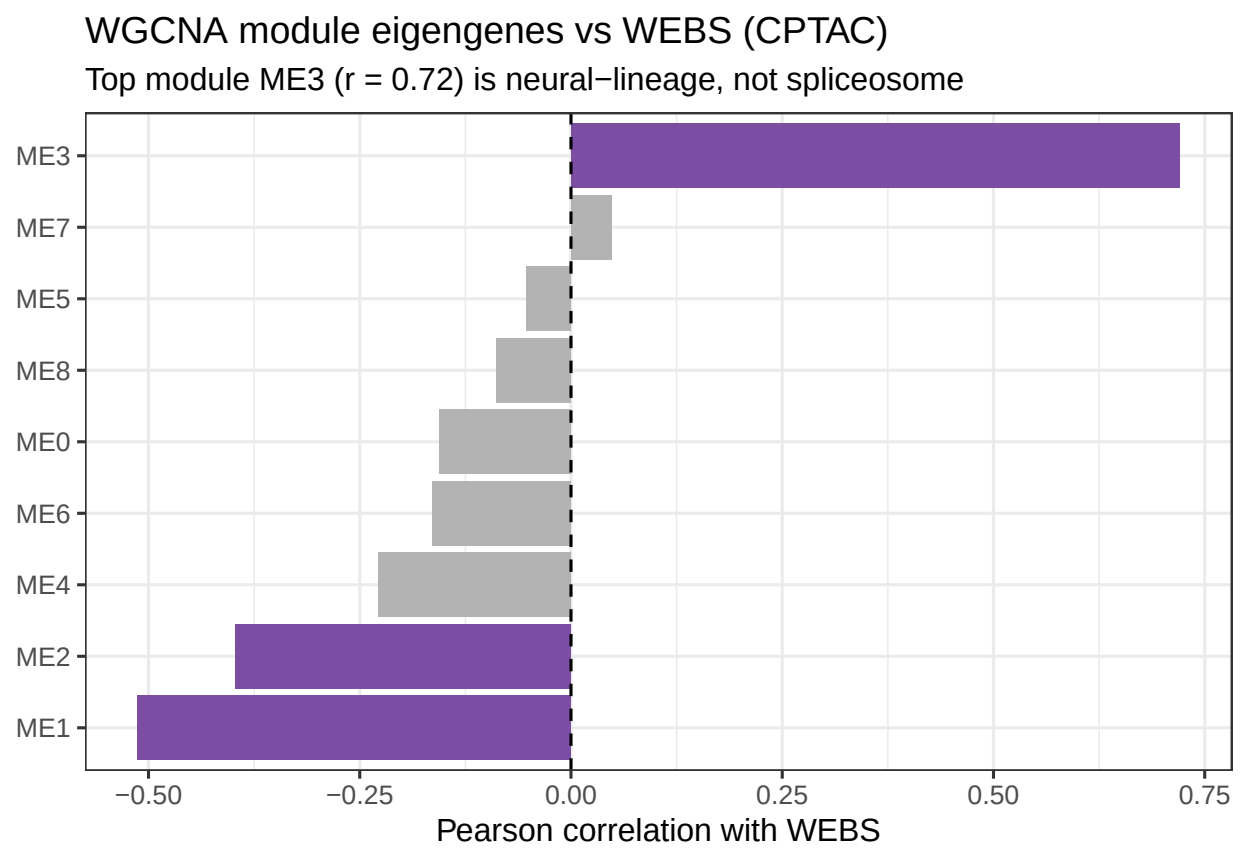

CGGA693: MGMT x signature joint stratification
up
joint=Methylated + High
joint=Methylated + Low
joint=Unmethylated + High
joint=Unmet
1.00
0.75
0.50
0.25
0.00
Overall survival
p < 0.0001
0
40
80
120
160
Time (months)


GLASS longitudinal (n=179 pairs, Wilcoxon p=0.091)
WEBS
2
1
0
−1
Primary
Recurrence

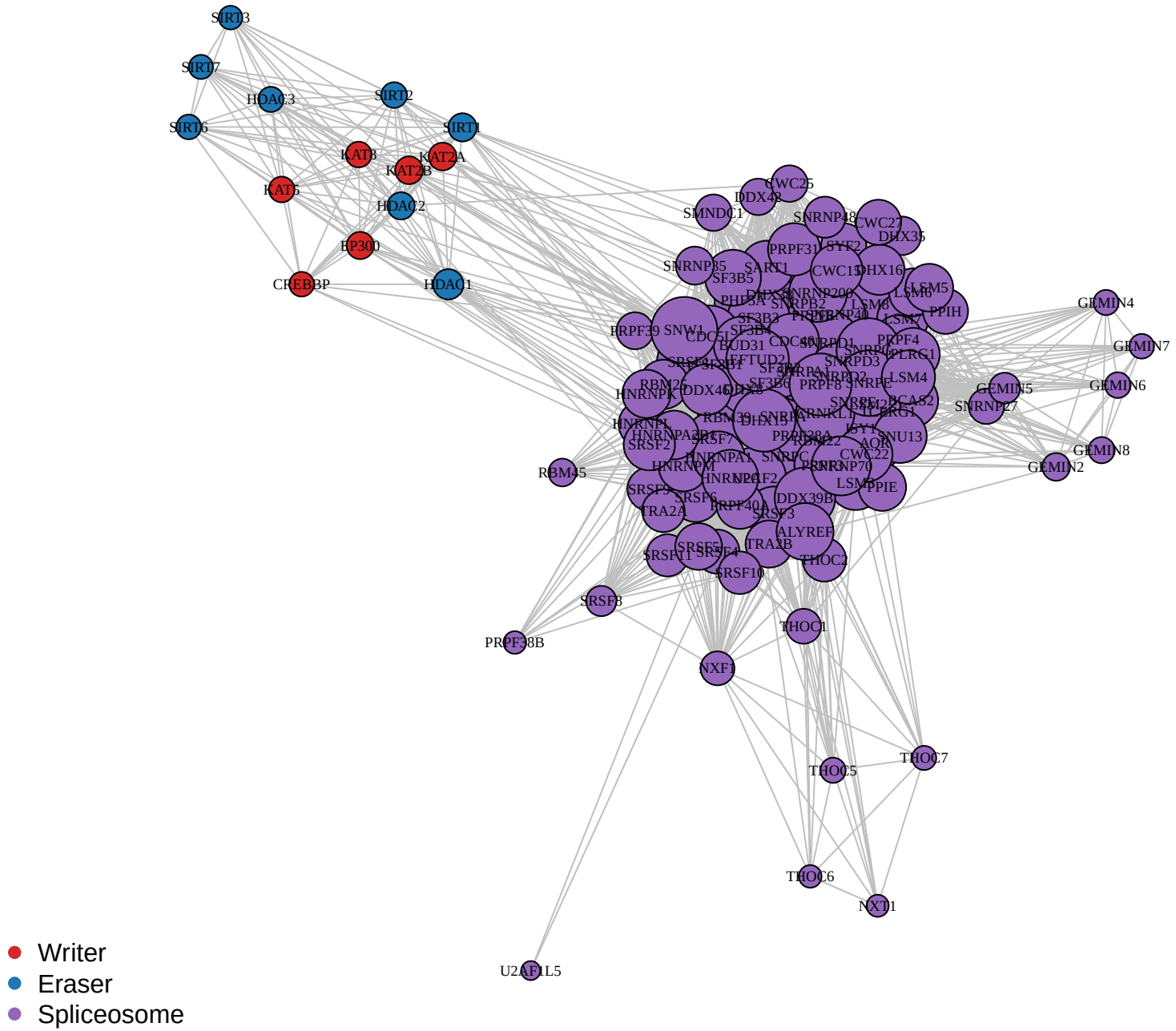

lnHR
SE(lnHR)
Hazard Ratio
HR
95% CI
Weight (common)
M
0.4776
0.1090
1.61
[1.30; 2.00]
4.7%
0.1291
0.0283
1.14
[1.08; 1.20]
69.5%
0.3517
0.0465
1.42
[1.30; 1.56]
25.8%
effect model
1.22
[1.17; 1.28]
100.0%
ffects model
1.35
[1.11; 1.65]
0.75
1
1.5
ity: I^2 = 91.4%, τ^2 = 0.0263, p < 0.0001
Hazard Ratio (per SD risk score)

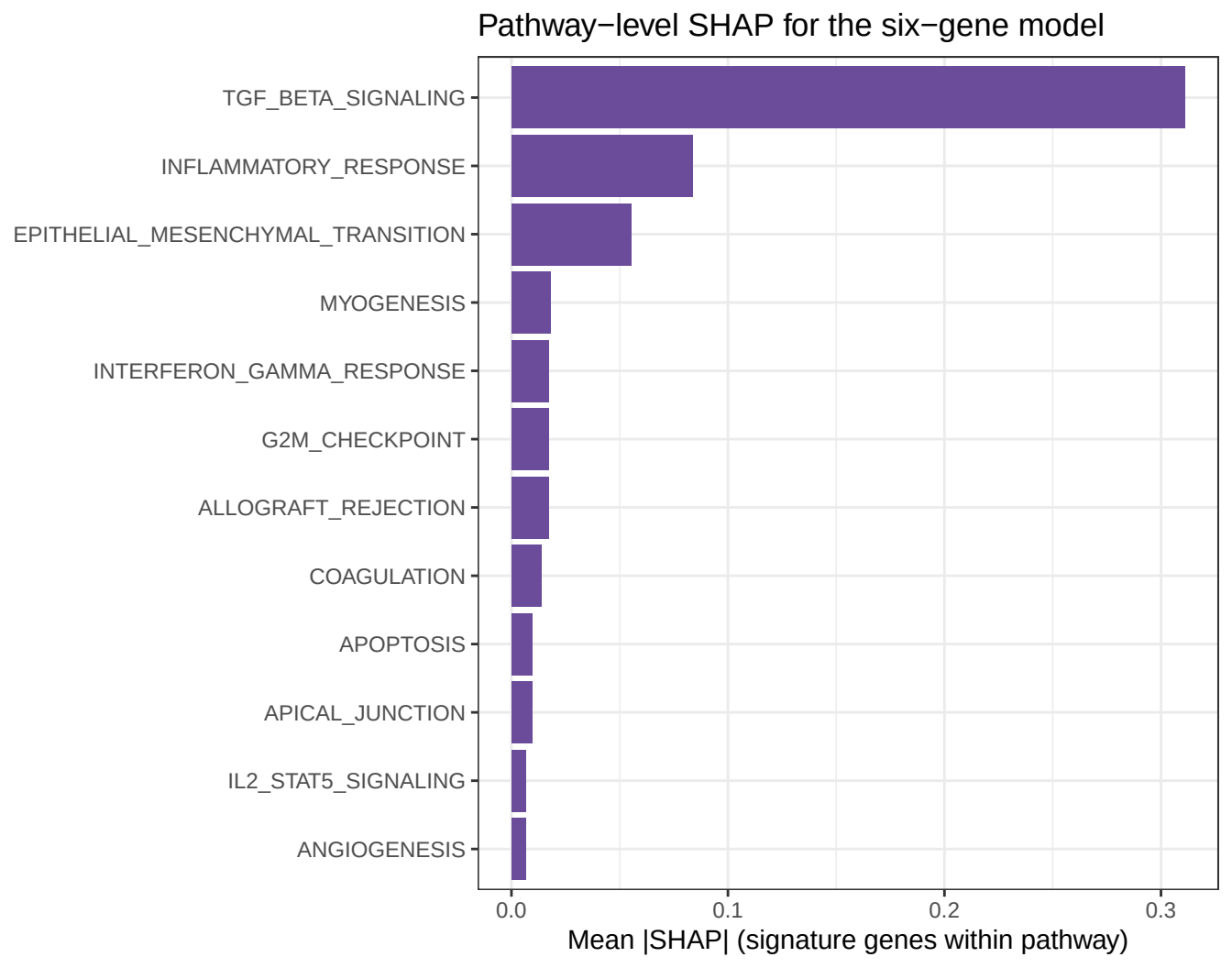

SNF fused similarity (k = 3, chi2 p = 4.1e−06)

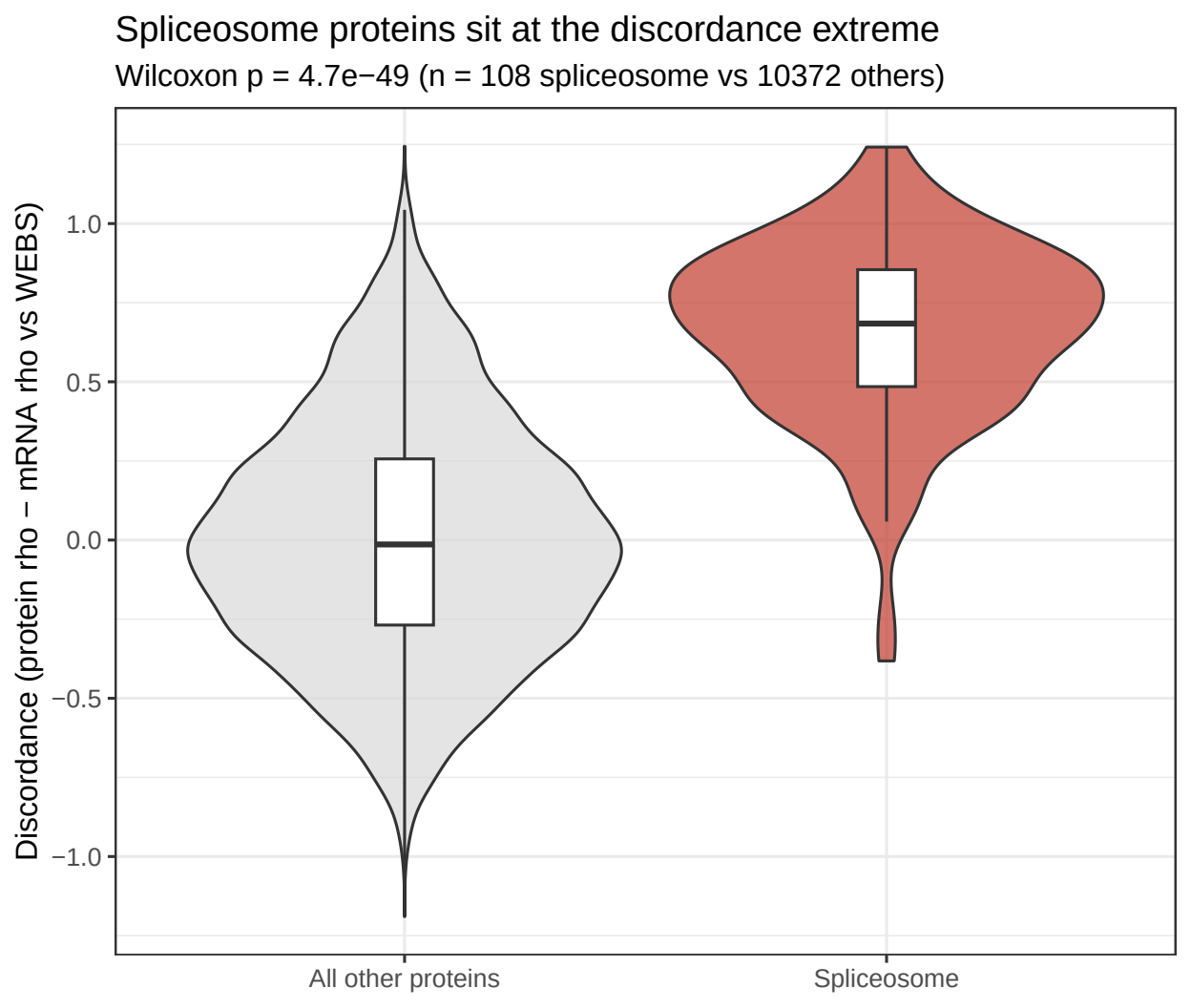

SNF consensus clusters vs WEBS (KW p = 7.1e−07)
Protein−level WEBS
0.4
0.0
−0.4
1
2
3
SNF cluster

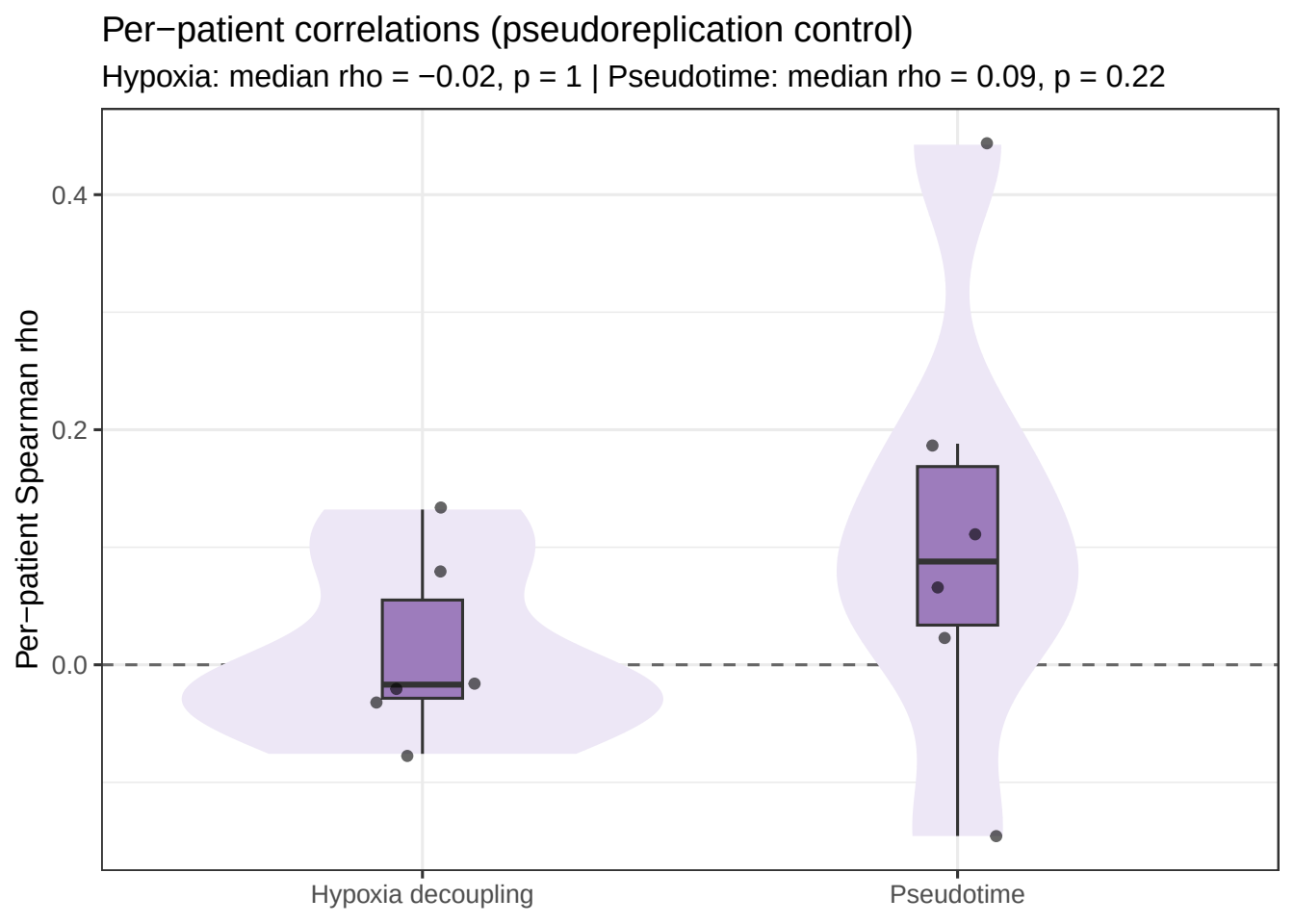

WGCNA module eigengenes vs WEBS (CPTAC)
Top module ME3 (r = 0.72) is neural−lineage, not spliceosome
ME3
ME7
ME5
ME8
ME0
ME6
ME4
ME2
ME1
−0.50
−0.25
0.00
0.25
0.50
0.75
Pearson correlation with WEBS

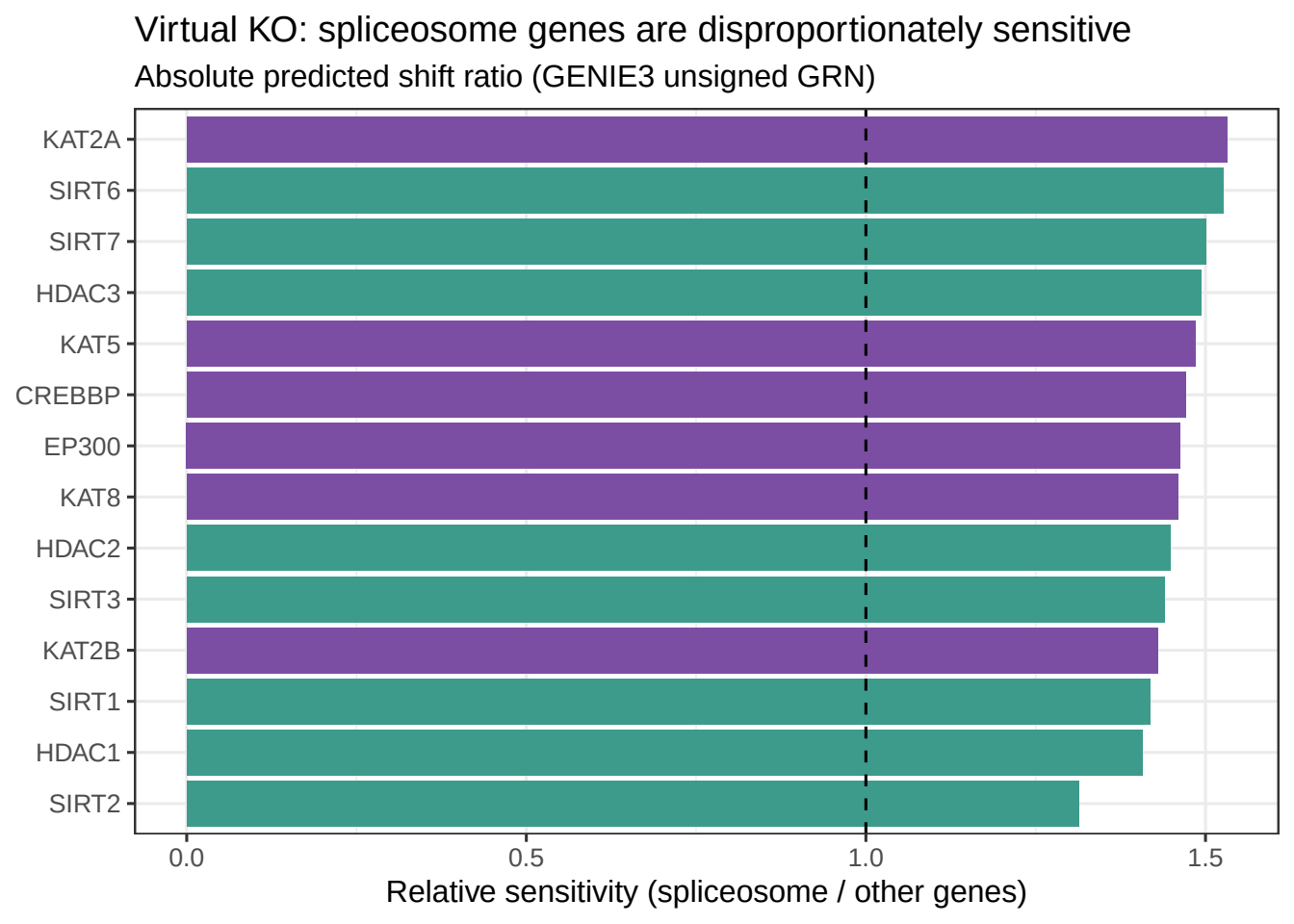
Writer−Eraser−Spliceosome PPI (121 nodes, 3349 edges)

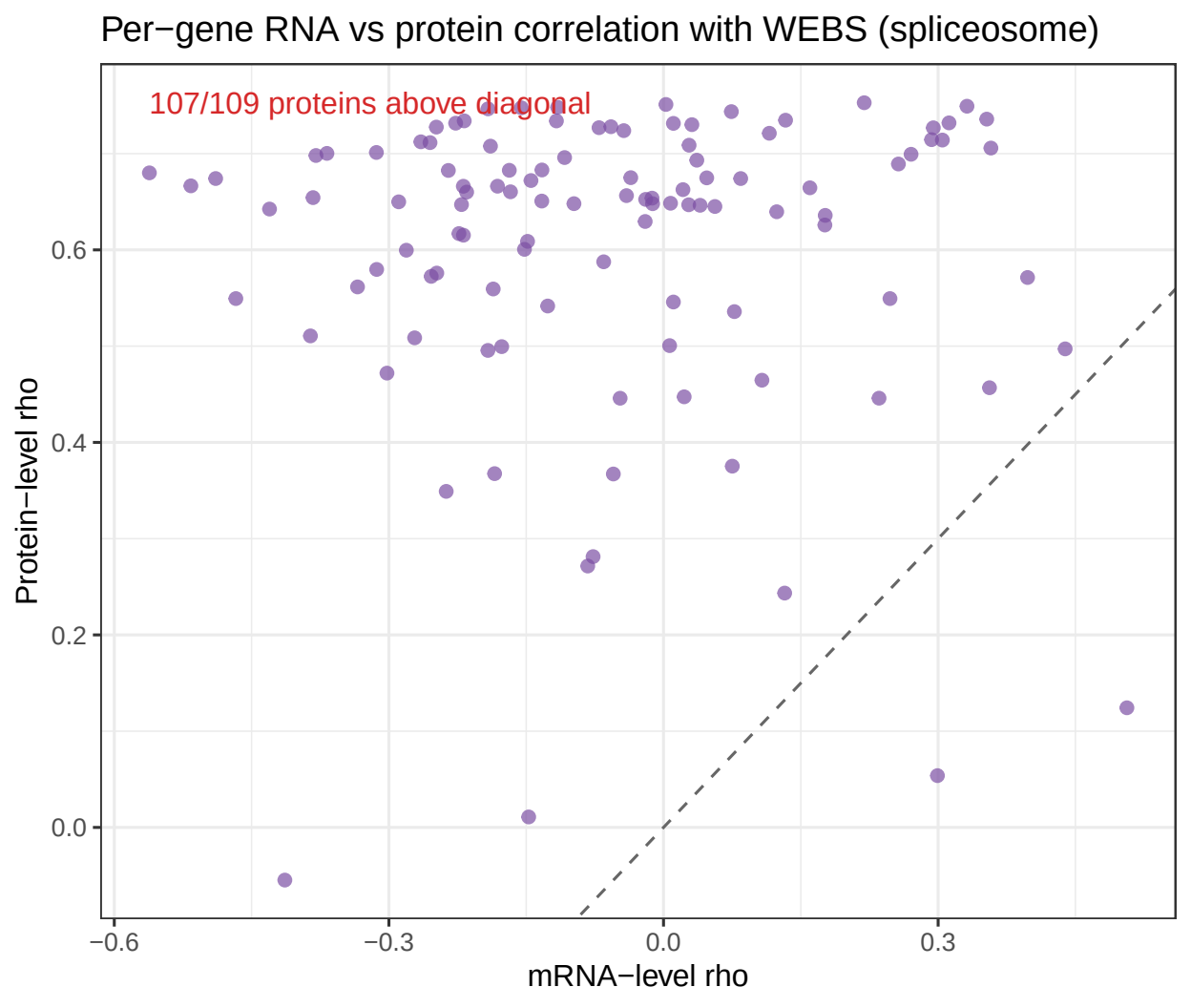


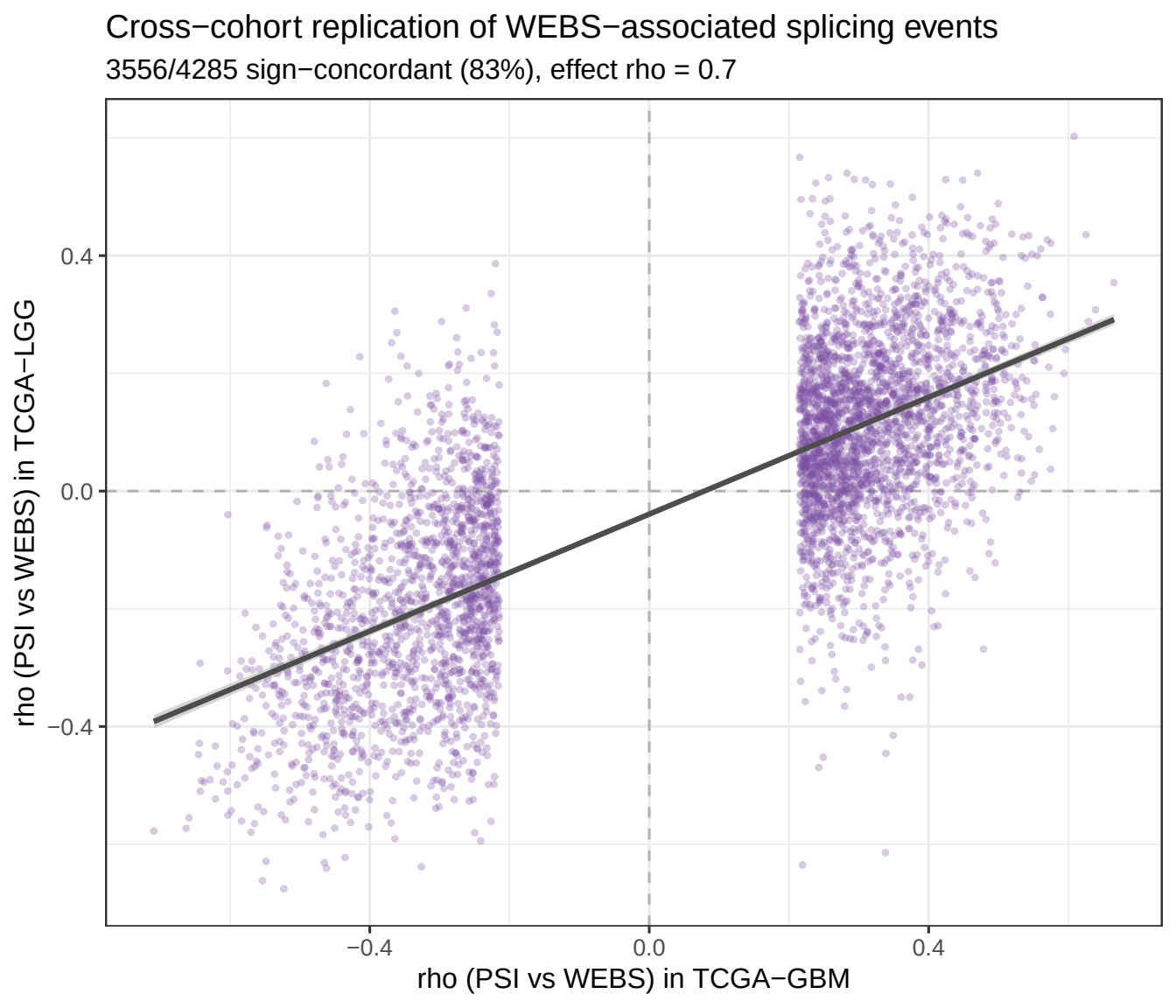

Pathway−level SHAP for the six−gene model
TGF_BETA_SIGNALING
INFLAMMATORY_RESPONSE
EPITHELIAL_MESENCHYMAL_TRANSITION
MYOGENESIS
INTERFERON_GAMMA_RESPONSE
G2M_CHECKPOINT
ALLOGRAFT_REJECTION
COAGULATION
APOPTOSIS
APICAL_JUNCTION
IL2_STAT5_SIGNALING
ANGIOGENESIS
0.0
0.1
0.2
0.3
Mean |SHAP| (signature genes within pathway)

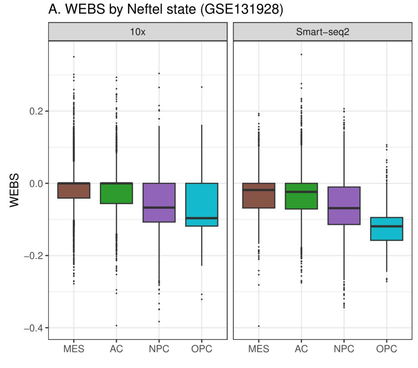

Spliceosome proteins sit at the discordance extreme
Wilcoxon p = 4.7e−49 (n = 108 spliceosome vs 10372 others)
Discordance (protein rho − mRNA rho vs WEBS)
1.0
0.5
0.0
−0.5
−1.0
All other proteins
Spliceosome

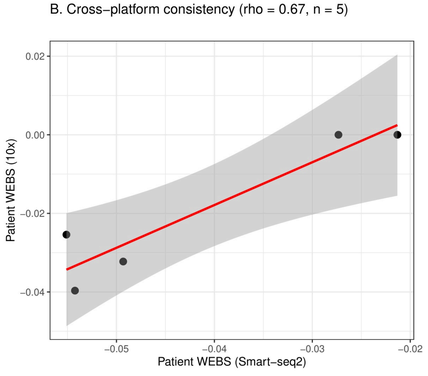

Per−patient correlations (pseudoreplication control)
Hypoxia: median rho = −0.02, p = 1 | Pseudotime: median rho = 0.09, p = 0.22
Per−patient Spearman rho
0.4
0.2
0.0
Hypoxia decoupling
Pseudotime

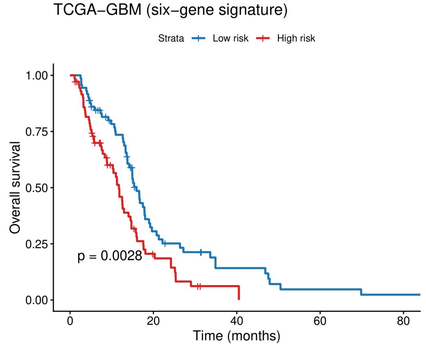

Virtual KO: spliceosome genes are disproportionately sensitive
Absolute predicted shift ratio (GENIE3 unsigned GRN)
KAT2A
SIRT6
SIRT7
HDAC3
KAT5
CREBBP
EP300
KAT8
HDAC2
SIRT3
KAT2B
SIRT1
HDAC1
SIRT2
0.0
0.5
1.0
1.5
Relative sensitivity (spliceosome / other genes)

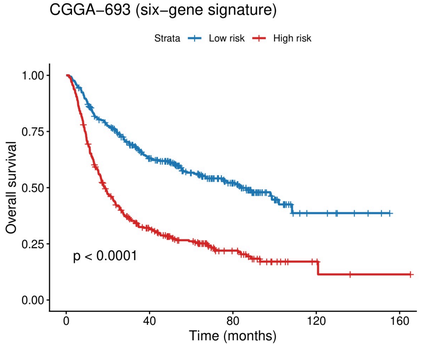

Per−gene RNA vs protein correlation with WEBS (spliceosome)
107/109 proteins above diagonal
Protein−level rho
mRNA−level rho
0.6
0.4
0.2
0.0
−0.6
−0.3
0.0
0.3

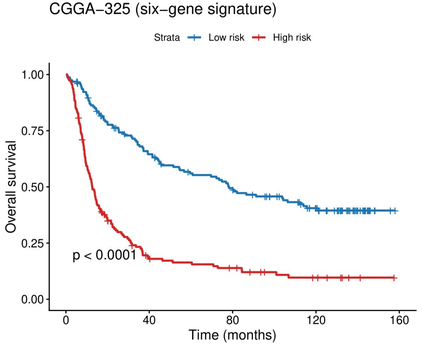

Cross−cohort replication of WEBS−associated splicing events
3556/4285 sign−concordant (83%), effect rho = 0.7
rho (PSI vs WEBS) in TCGA−LGG
rho (PSI vs WEBS) in TCGA−GBM
0.4
0.0
−0.4
−0.4
0.0
0.4

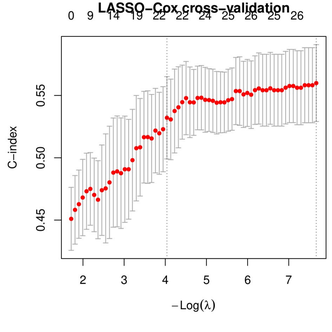

A. WEBS by Neftel state (GSE131928)
10x
Smart−seq2
WEBS
0.2
0.0
−0.2
−0.4
MES
AC
NPC
OPC
MES
AC
NPC
OPC

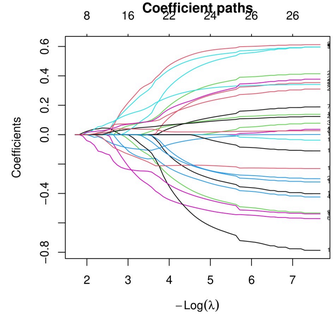

B. Cross−platform consistency (rho = 0.67, n = 5)
Patient WEBS (10x)
0.02
0.00
−0.02
−0.04
−0.05
−0.04
−0.03
−0.02
Patient WEBS (Smart−seq2)

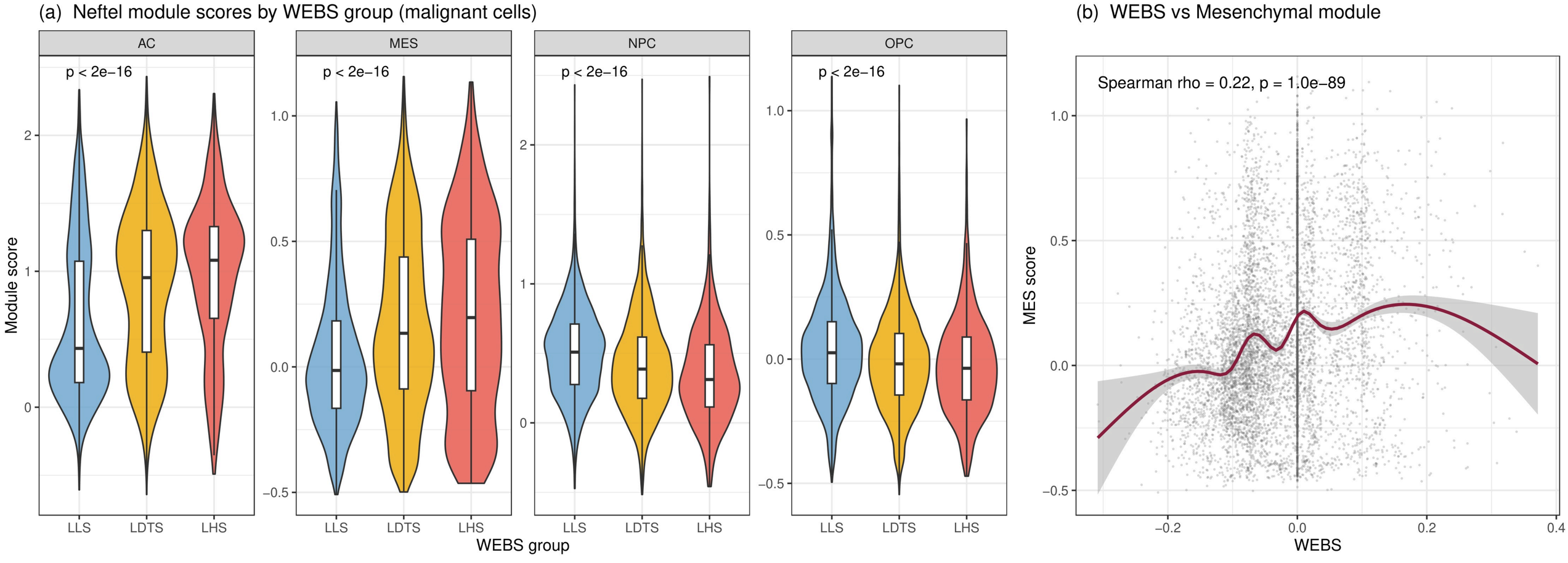

TCGA−GBM (six−gene signature)
Strata
Low risk
High risk
Overall survival
1.00
0.75
0.50
0.25
0.00
p = 0.0028
0
20
40
60
80
Time (months)


CGGA−693 (six−gene signature)
Strata
Low risk
High risk
Overall survival
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
40
80
120
160
Time (months)


CGGA−325 (six−gene signature)
Strata
Low risk
High risk
Overall survival
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
40
80
120
160
Time (months)

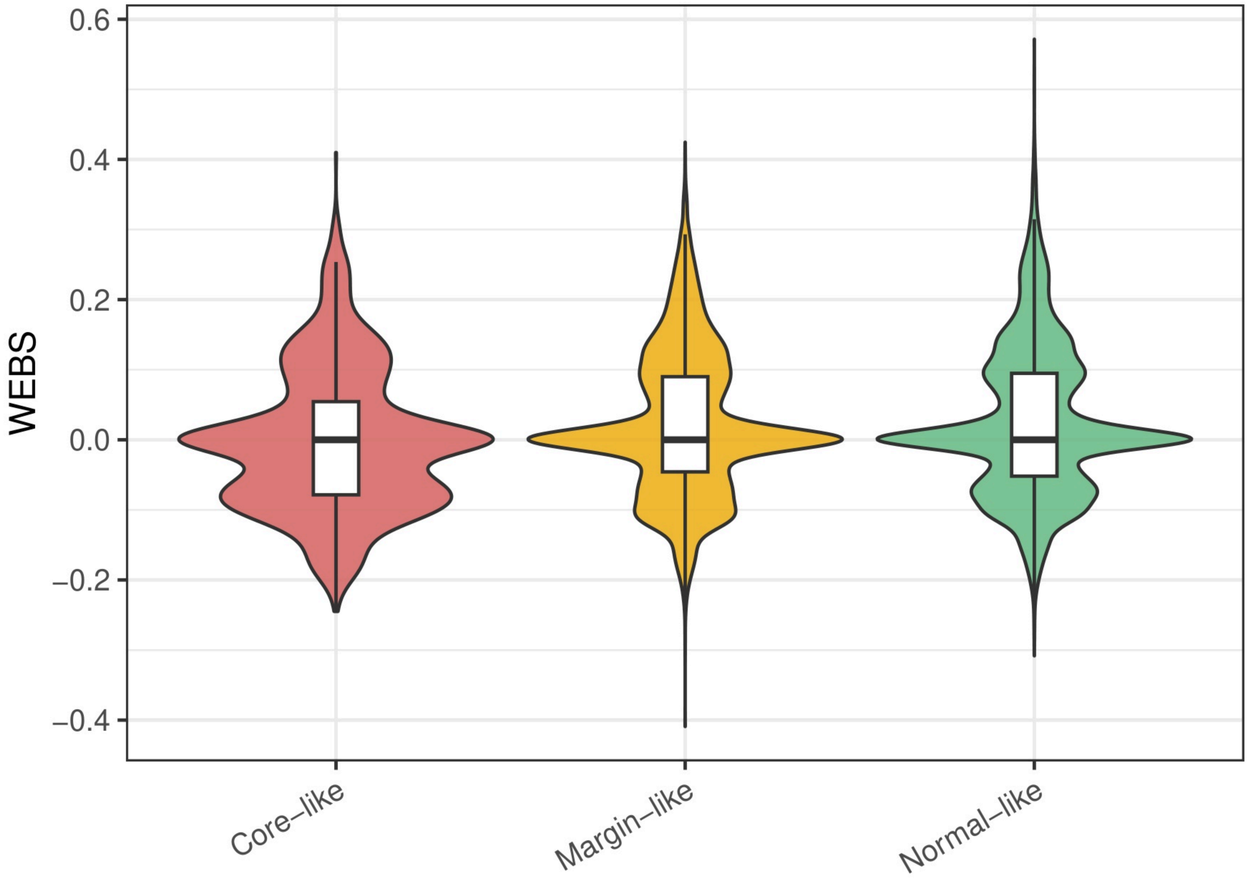

LASSO−Cox cross−validation
0 9 14 19 22 22 24 25 26 25 26
C−index
0.55
0.50
0.45
2
3
4
5
6
7
−Log(λ)


Coefficient paths
8 16 22 24 26 26
Coefficients
0.6
0.4
0.2
0.0
−0.4
−0.8
2
3
4
5
6
7
−Log(λ)


(a) Neftel module scores by WEBS group (malignant cells)
AC
MES
NPC
OPC
p < 2e−16
Module score
LLS
LDTS
LHS
WEBS group
(b) WEBS vs Mesenchymal module
Spearman rho = 0.22, p = 1.0e−89
MES score
WEBS

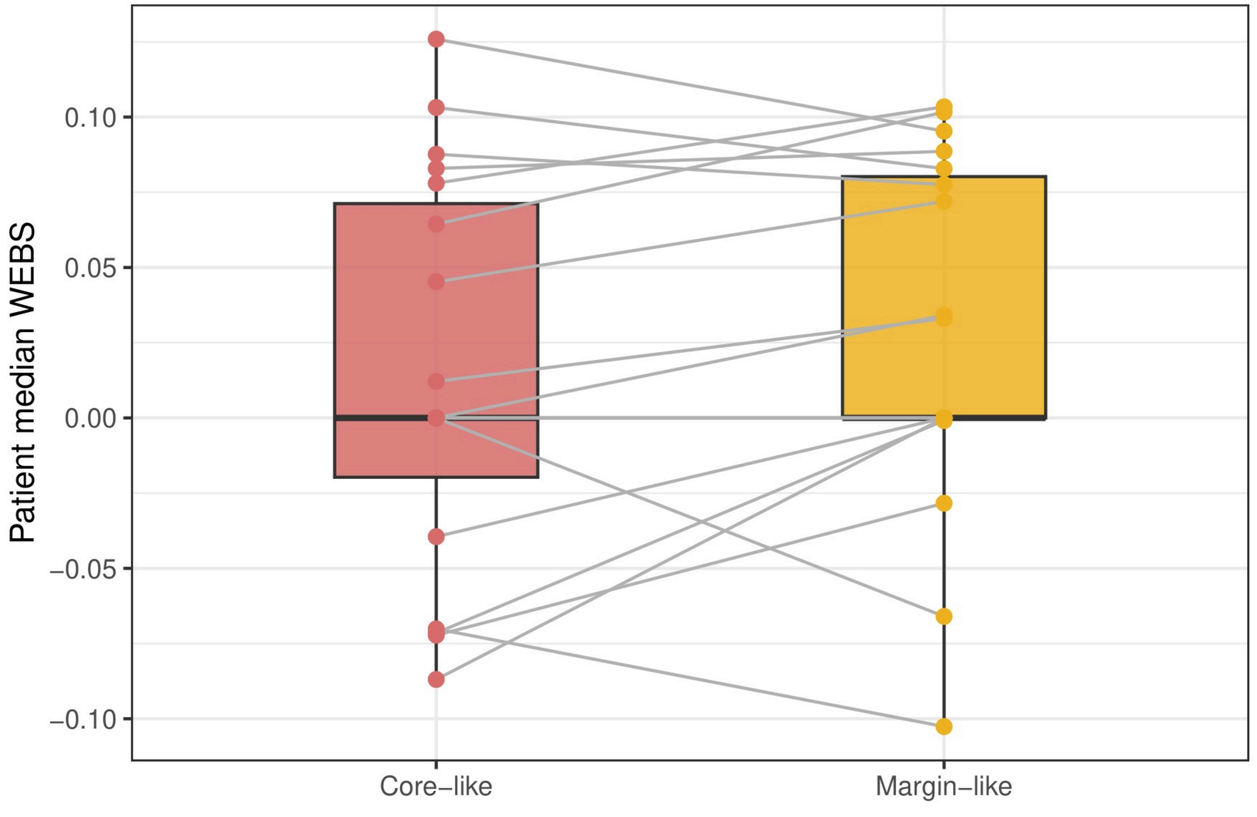
(a) WEBS across invasion region types

Kruskal–Wallis p = 7.4e–25



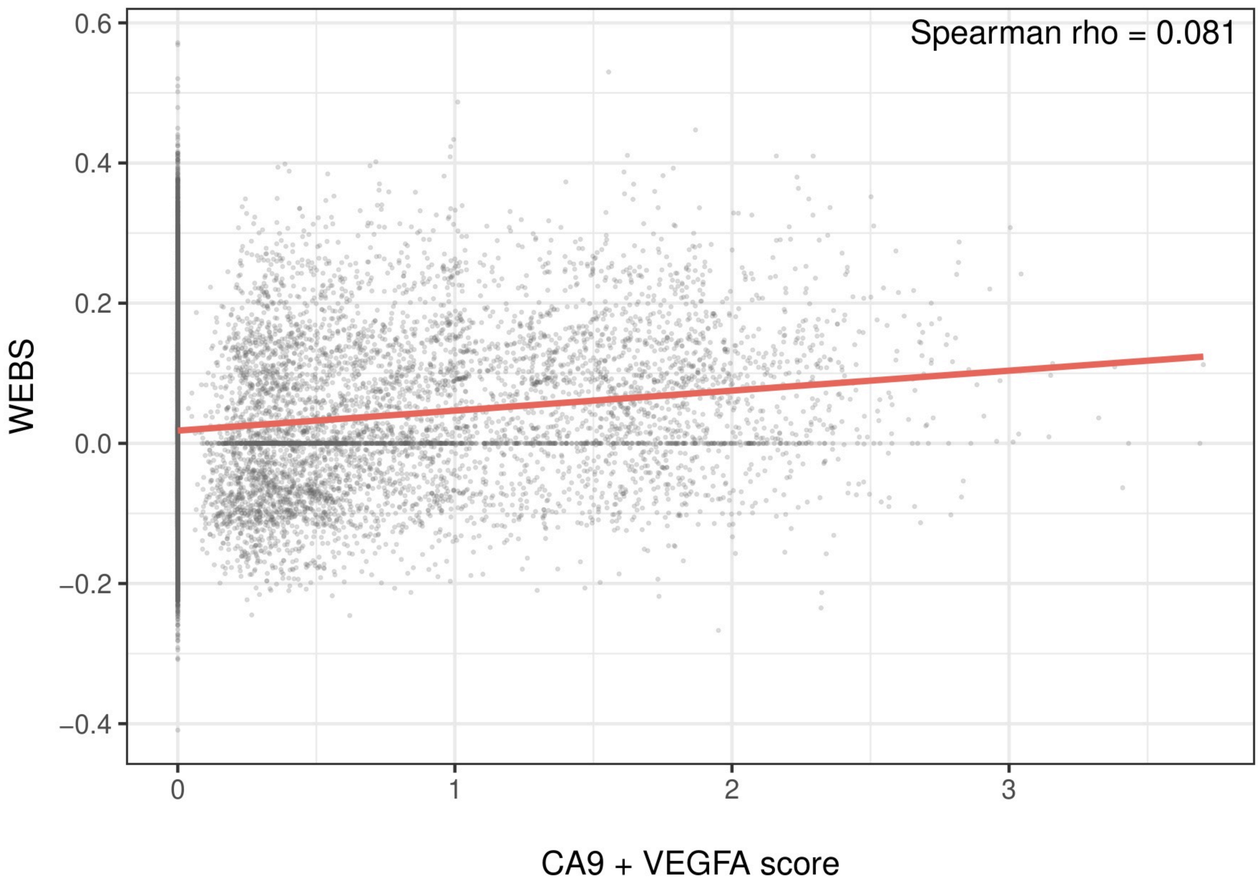
(b) Patient−level paired Margin vs Core WEBS

paired Wilcoxon, n=19, p=1.3e−01



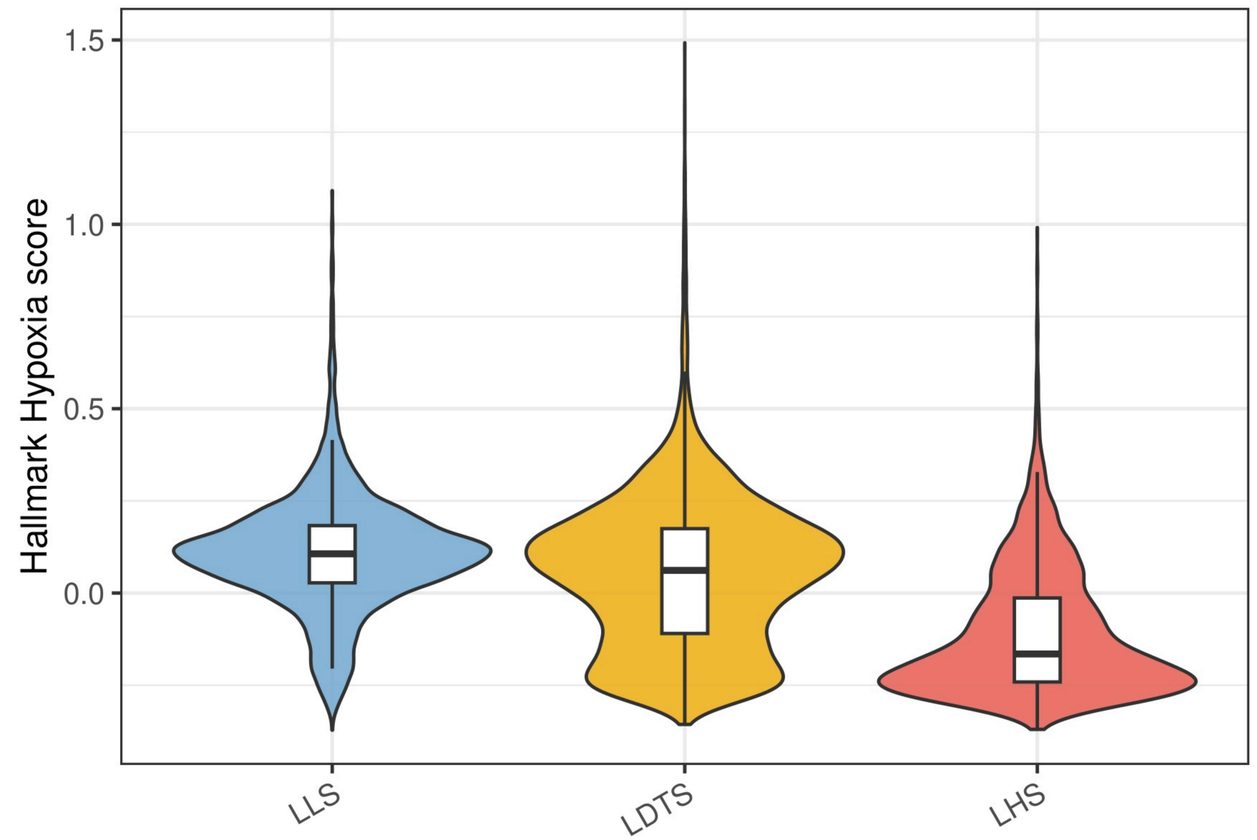
(c) CA9 + VEGFA vs WEBS

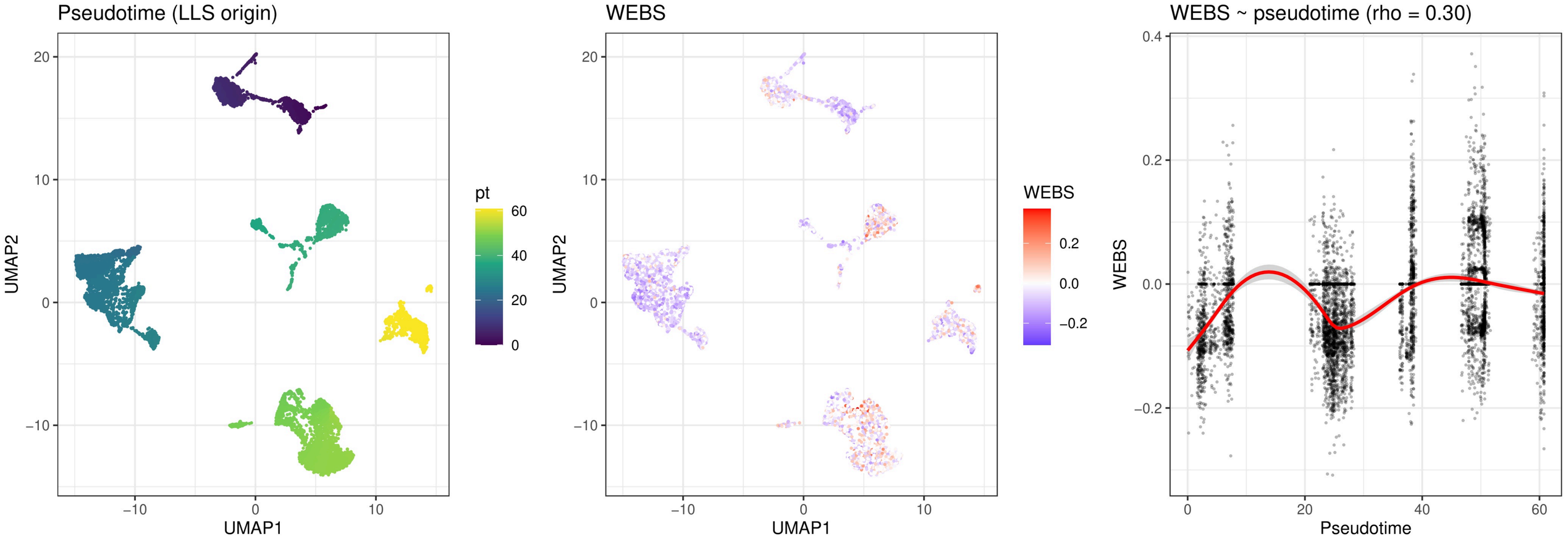
All cells (n = 35,823)

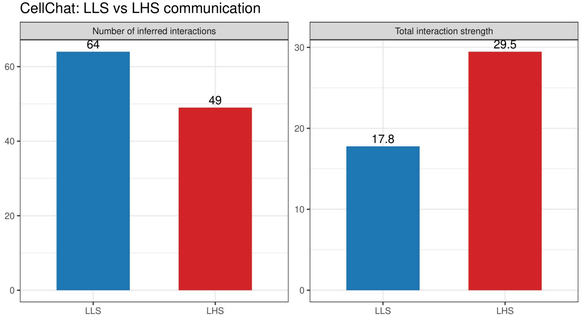


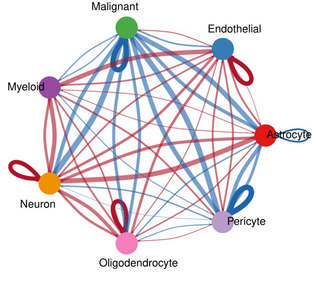
(d) Hallmark Hypoxia score by WEBS group

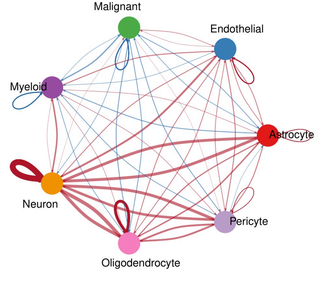
p < 1e−300



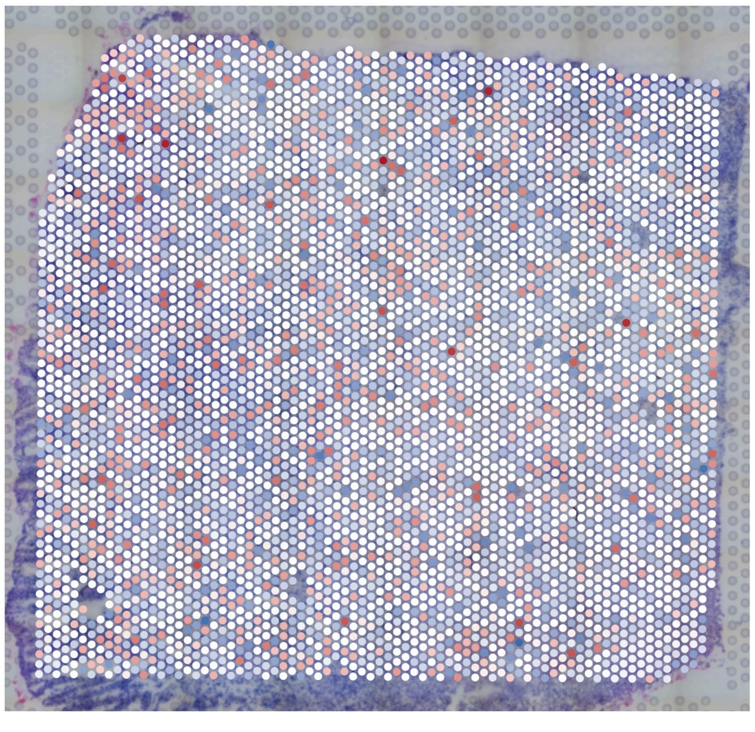

Pseudotime (LLS origin)
UMAP2
UMAP1
pt
60
40
20
0
WEBS
WEBS
0.2
0.0
−0.2
WEBS ~ pseudotime (rho = 0.30)
WEBS
Pseudotime


CellChat: LLS vs LHS communication
Number of inferred interactions
64
49
60
40
20
0
LLS
LHS
Total interaction strength
29.5
17.8
30
20
10
0
LLS
LHS

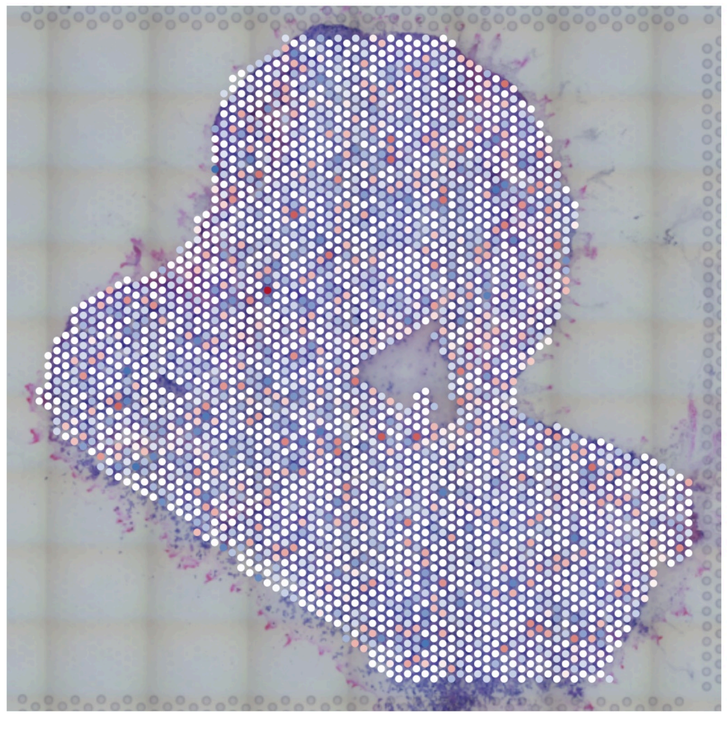

Malignant
Endothelial
Myeloid
Astrocyte
Neuron
Pericyte
Oligodendrocyte


Malignant
Endothelial
Myeloid
Astrocyte
Neuron
Pericyte
Oligodendrocyte

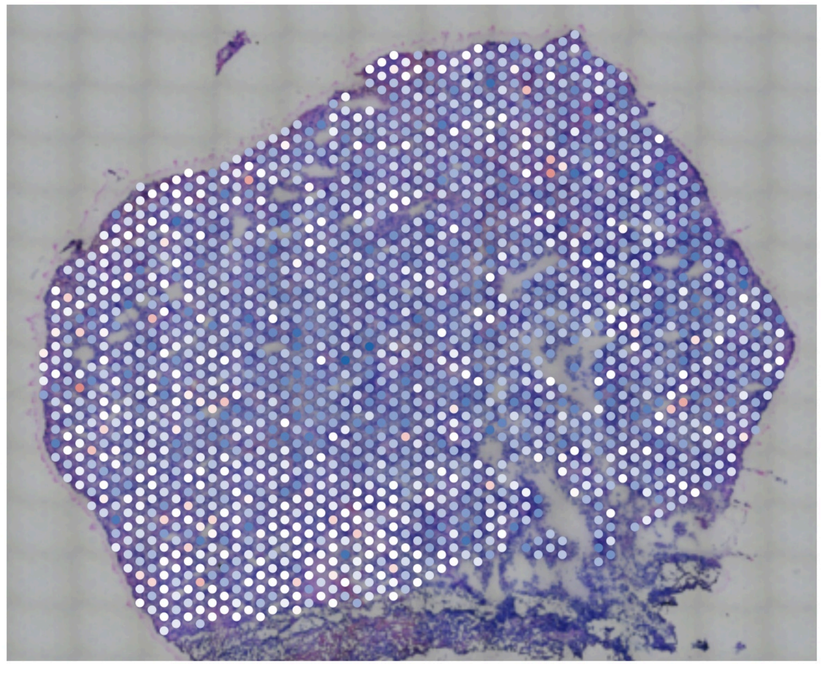
GBM1 WEBS



GBM2 WEBS



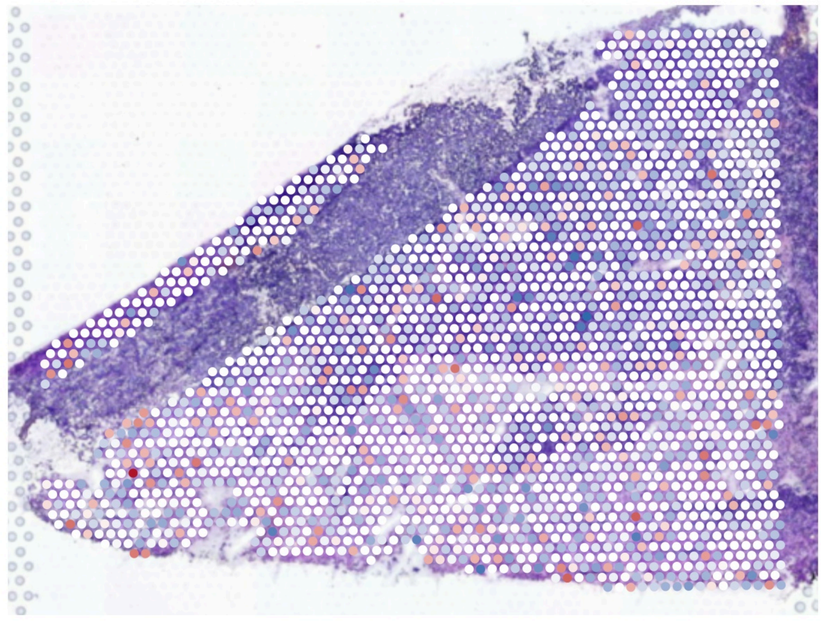
GBM3 WEBS



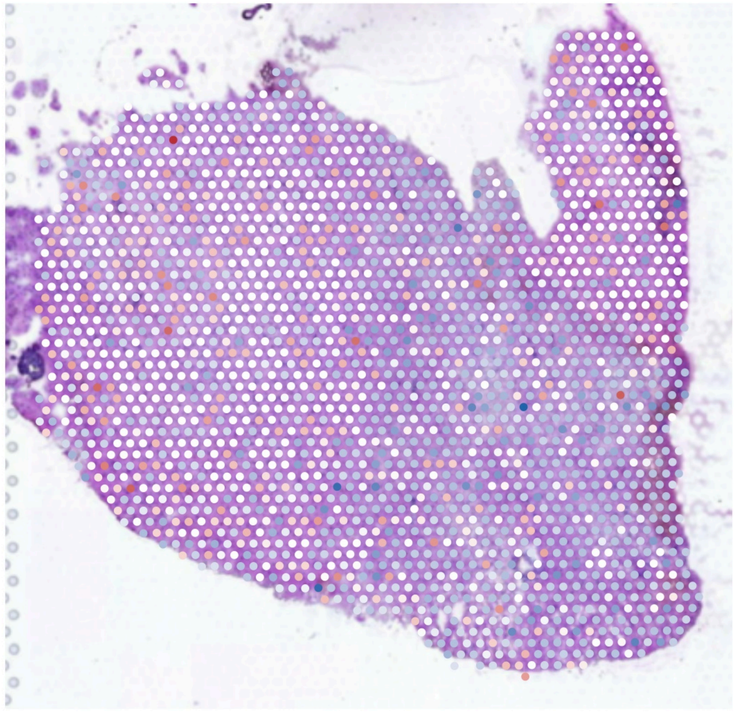
GBM4 WEBS



GBM5_1 WEBS



GBM5_2 WEBS



snATAC−seq promoter accessibility (TSS +/− 2kb) at writer/eraser loci

log1p(peak counts)

0
1
2
3

EP300
HDAC1
HDAC2
KAT2B
KAT5
KAT8
SIRT1
SIRT2
SIRT3
SIRT6

Eraser
Writer

B. Promoter accessibility vs patient WEBS (6 snATAC samples)

Mean promoter accessibility (log1p)

EP300
HDAC1
HDAC2
KAT2B
KAT5
KAT8
SIRT1
SIRT2
SIRT3
SIRT6

Patient median WEBS (scRNA−seq)

Eraser
Writer

(a) Mesenchymal score by WEBS group

Kruskal–Wallis p = 2.4e−67

Neftel Mesenchymal Score

1.0
0.5
0.0
−0.5

LLS
LDTS
LHS

(b) AUCell pathway activity (LHS – LLS)

EPITHELIAL_MESENCHYMAL_TRANSITION
INFLAMMATORY_RESPONSE
APOPTOSIS
HYPOXIA
GLYCOLYSIS
PI3K_AKT_MTOR_SIGNALING
G2M_CHECKPOINT
TNFA_SIGNALING_VIA_NFKB
MTORC1_SIGNALING
E2F_TARGETS
DNA_REPAIR
UNFOLDED_PROTEIN_RESPONSE
MYC_TARGETS_V2
MYC_TARGETS_V1

−0.06
−0.04
−0.02
0.00

AUC Diff (LHS−LLS)

(c) Stemness by WEBS group

Kruskal–Wallis p = 1.0e−184

Stemness Score

2
1
0
−1
−2

LLS
LDTS
LHS

(d) Perturbation sensitivity of WEBS (leave−one−out)

SIRT3
SIRT6
SIRT7
SIRT1
HDAC1
HDAC3
KAT8
KAT5
KAT2A
HDAC2
EP300
CREBBP
KAT2B
SIRT2

−0.09
−0.06
−0.03
0.00

Delta correlation vs default

Type
Eraser
Writer

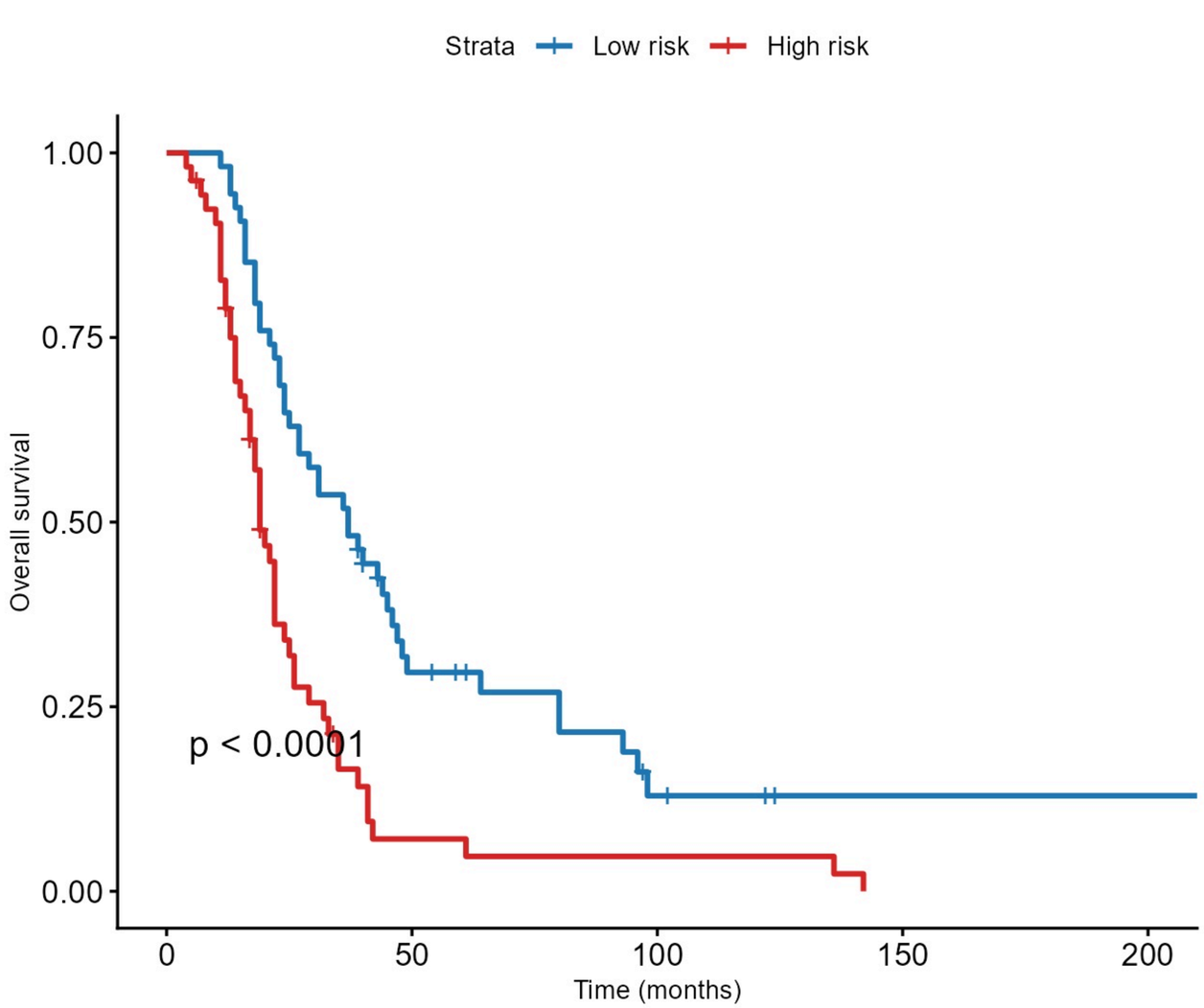
(e) Top differential TFs (FDR < 0.05)

ZEB1
ID2
FOSB
ATF4
TCF12
ASCL1
EGR1
JUNB
SOX2
JUN
FOS
OLIG2

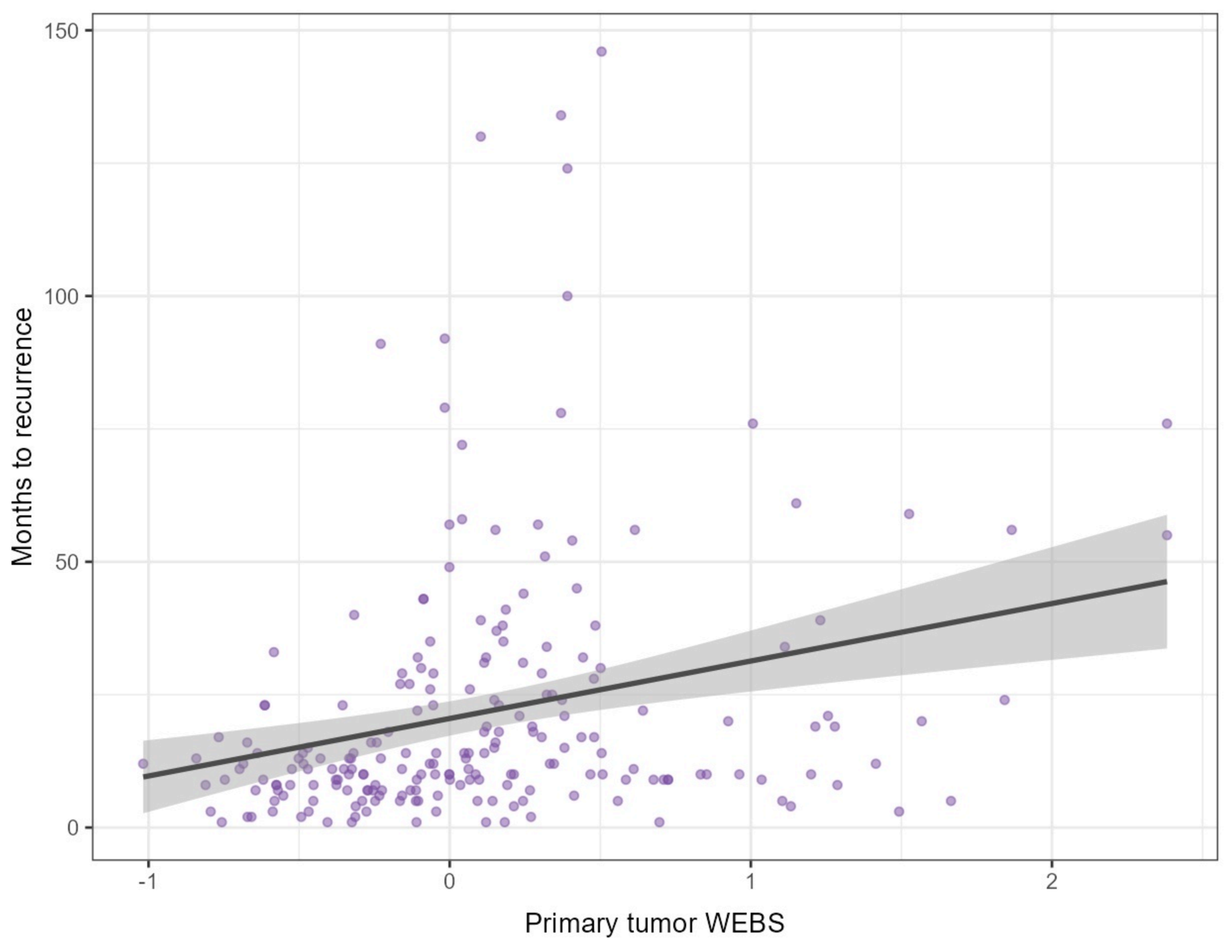
−0.75
−0.50
−0.25
0.00

log2 Fold Change (LHS vs LLS)

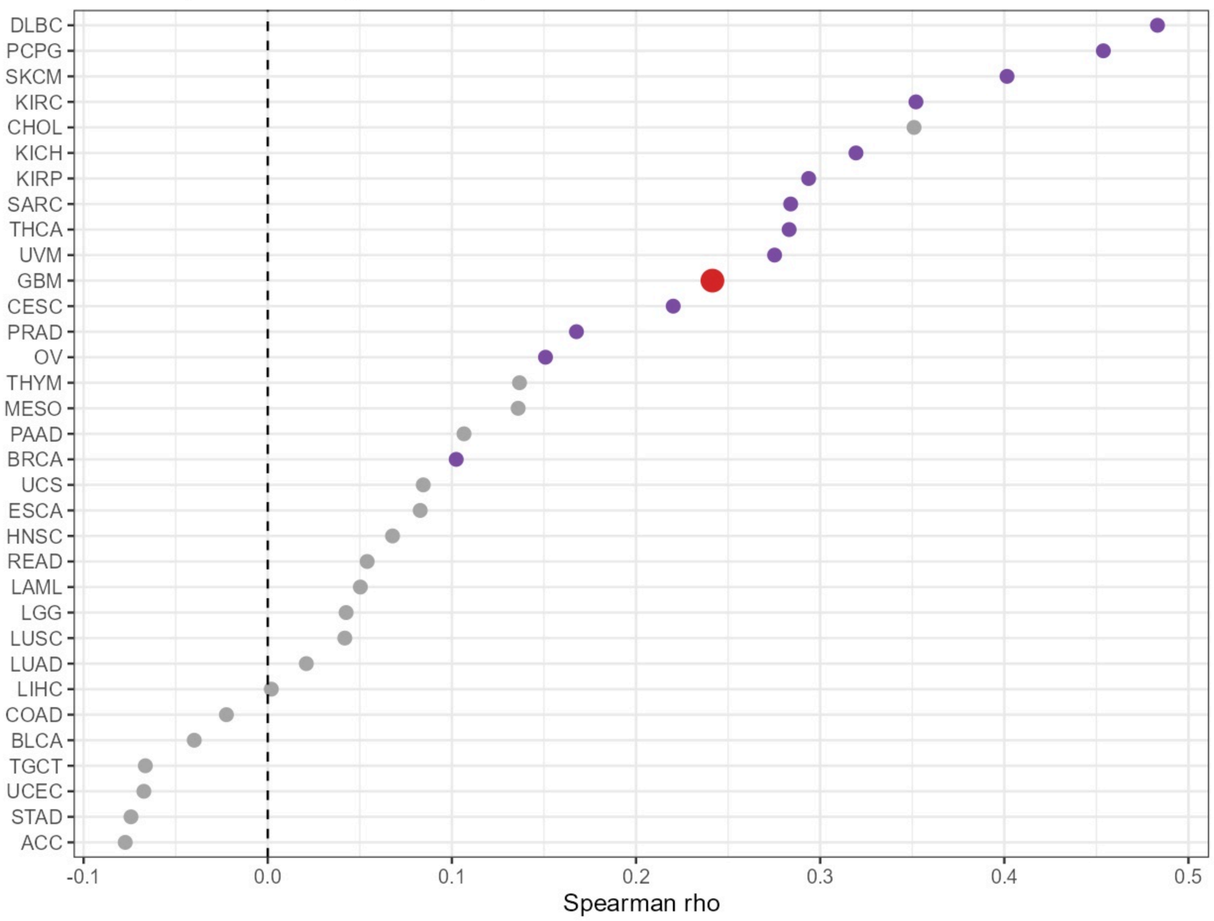
a


GLASS primary (n=107)
Strata
Low risk
High risk
Overall survival
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
50
100
150
200
Time (months)

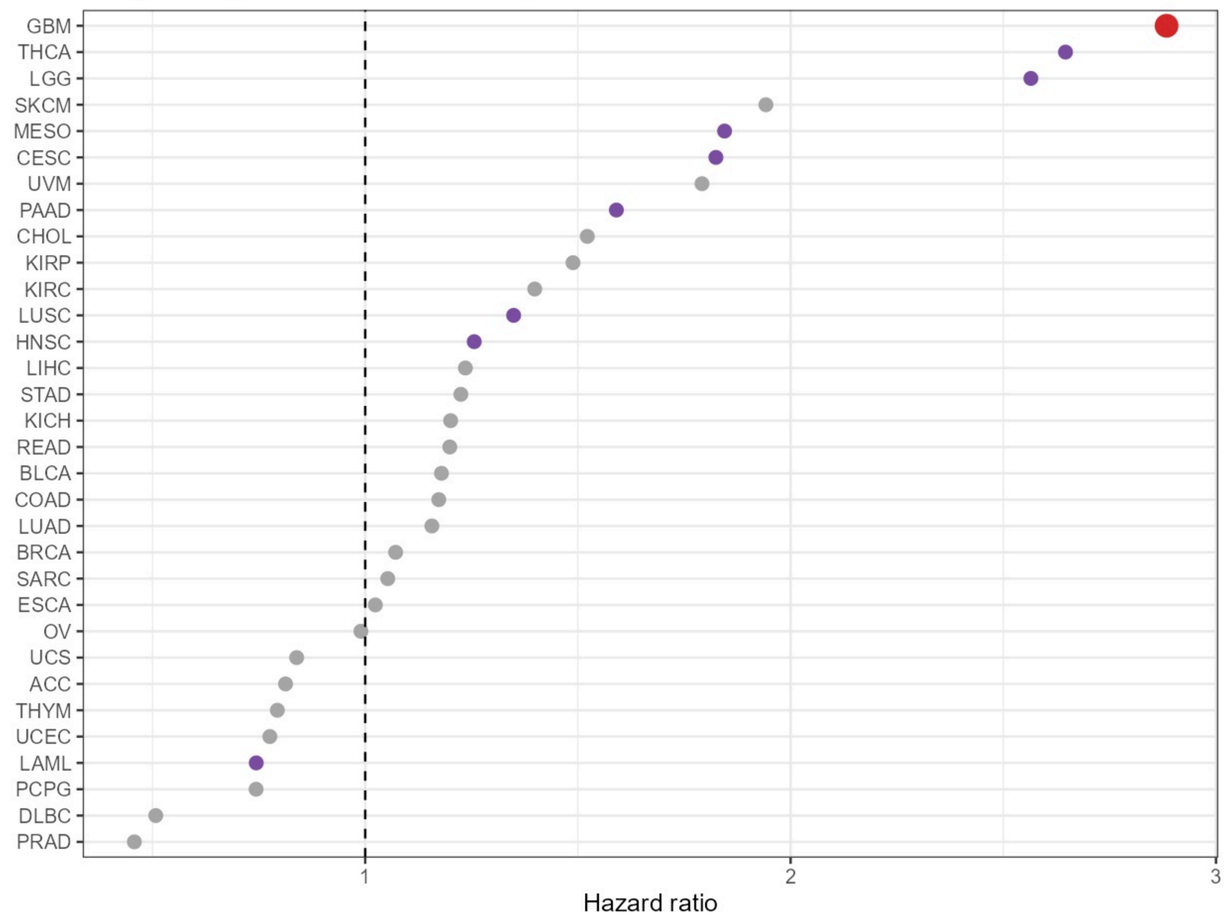
b


GLASS recurrence (rho=0.33, p=1e-06)
Months to recurrence
150
100
50
0
-1
0
1
2
Primary tumor WEBS

c


WEBS-spliceosome (33 TCGA)
DLBC
PCPG
SKCM
KIRC
CHOL
KICH
KIRP
SARC
THCA
UVM
GBM
CESC
PRAD
OV
THYM
MESO
PAAD
BRCA
UCS
ESCA
HNSC
READ
LAML
LGG
LUSC
LUAD
LIHC
COAD
BLCA
TGCT
UCEC
STAD
ACC
-0.1
0.0
0.1
0.2
0.3
0.4
0.5
Spearman rho

d


Six-gene signature HR (33 TCGA)
GBM
THCA
LGG
SKCM
MESO
CESC
UVM
PAAD
CHOL
KIRP
KIRC
LUSC
HNSC
LIHC
STAD
KICH
READ
BLCA
COAD
LUAD
BRCA
SARC
ESCA
OV
UCS
ACC
THYM
UCEC
LAML
PCPG
DLBC
PRAD
1
2
3
Hazard ratio

Acetylation increases

Writers:
EP300, KAT2A/B,
KAT5, KAT8, CREBBP

Erasers:
HDAC1-3,
SIRT1-3,6,7

Deacetylation decreases

Writer-dominant state (LHS)

Non-histone acetyltransferase activity

Splicing-factor lysine acetylation (43 sites; OR=7.36)

Blocks ubiquitination (competition)

Ub

Protein stabilization ($p = 0.62$)

SPLICEOSOME ACCUMULATION

107/111 proteins up-regulated · $p = 3.4 \times 10^{-37}$ · protein $p = +0.79$ · mRNA $p = -0.03$

U1 U4/U6 U5

Spliceosome complex
- snRNPs
- SF3B
- PRP19 complex
- NTC

Enhanced DNA repair

Ribosome biogenesis

Survival & proliferation

Immunosuppressive milieu

MIF

CD74

HDAC inhibitors

vorinostat • entinostat
HDAC2 docking – 7.9 kcal/mol

Spliceosome modulators

H3B-8800 • E7107
SF3B1 docking – 6.7 kcal/mol

Synergistic targeting